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College of Information Technology Education

**A Web-Based Integrated School Management System for Timothy's Cradle Christian
International School Inc.**

In Partial Fulfillment of the Requirements
For Capstone 2 and Research Subject

Elgie P. Delos Reyes

Carl John M. Cero

Ma. Mediatrix T. Fruto

Kylen Marie Sta. Maria

Eunice M. Lagunday

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CHAPTER I

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The rapid development of information and communication technologies has significantly changed the way organizations manage information, perform daily operations, and communicate with their stakeholders. In the education sector, digital technologies have become increasingly important in improving administrative processes, academic management, communication, and information accessibility. However, despite the growing adoption of digital systems, many educational institutions, particularly small private schools and schools located in underserved areas, continue to rely heavily on manual and paper-based processes for managing student records, attendance, inventory, reports, and stakeholder communication [1]. These traditional methods are often maintained because of limited financial resources, lack of technical expertise, inadequate infrastructure, and uncertainty regarding the complexity of implementing and maintaining computerized systems [2].

Manual information management requires school personnel to perform repetitive tasks such as writing records, updating logbooks, organizing documents, calculating grades, preparing attendance reports, monitoring inventory, and communicating important announcements to parents and teachers. Although these methods may be manageable when the number of students and school activities is small, they become increasingly difficult to maintain as the school grows. The larger the volume of information, the more time and effort are required to organize, retrieve, update, and maintain records. Consequently, manual systems may result in delays, duplication of information, inconsistent records, human errors, misplaced documents, and difficulties in monitoring important school activities [3].

Student records are among the most important types of information maintained by an educational institution. These records may include personal information, enrollment details, parent or guardian information, attendance history, academic performance, and other relevant student data. When these records are maintained primarily through physical documents and separate spreadsheets, retrieving specific information can become time-consuming. School personnel may need to search through filing cabinets, folders, notebooks, or multiple computer files before finding the required information. This process can affect the efficiency of administrative operations and may also increase the possibility of records being misplaced, damaged, duplicated, or accidentally modified. A centralized digital system can help address these concerns by allowing authorized users to store and retrieve information through a structured database.

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Attendance monitoring is another important school activity that can be affected by manual processes. Teachers traditionally record student attendance using attendance sheets, class lists, or logbooks. At the end of a particular period, attendance information may need to be reviewed and summarized for administrative reporting. When this information is recorded manually, teachers and school staff may spend considerable time checking records and preparing summaries. Errors may also occur when attendance information is transferred from one document to another. A computerized attendance module can provide a more organized method for recording daily attendance and maintaining historical attendance information. It can also make it easier for authorized personnel to review attendance records and generate reports when needed.

Inventory management is similarly important in ensuring that school resources and materials are properly monitored. Educational institutions regularly use classroom supplies, learning materials, office equipment, books, furniture, and other school resources. In a manual inventory system, information about available items, quantities, borrowed materials, and returned items may be recorded in notebooks, forms, or spreadsheets. Without a centralized monitoring mechanism, it can be difficult to determine the current quantity of supplies or identify which materials are currently being used or borrowed. This may result in inaccurate inventory records, delayed replenishment, or difficulty tracking school property. A digital inventory management component can provide school personnel with a more systematic way of recording, updating, and monitoring school resources.

Communication between parents, teachers, and school administrators is also an essential part of effective school management. Parents need timely information regarding their children's attendance, academic performance, school activities, announcements, and other important concerns. Teachers and administrators, on the other hand, need an efficient way to communicate information and respond to parent inquiries. When communication depends mainly on printed notices, personal messages, or separate communication channels, important information may be delayed, overlooked, or difficult to retrieve. A centralized communication portal can provide an organized platform where authorized users can share announcements and relevant information, helping strengthen communication between the school and the families it serves.

At Timothy's Cradle Christian International School Inc. (TCCIS), administrative and academic workflows such as enrollment, attendance tracking, grading, reporting, inventory monitoring, and stakeholder communication are primarily managed through manual processes involving physical files, logbooks, and basic spreadsheets. While these methods provide a familiar way of managing school operations, they become increasingly challenging as the volume of school information and activities increases. As enrollment grows and administrative responsibilities expand, school personnel need to manage more records and perform more repetitive tasks. This creates a need for a system that can organize information more efficiently while remaining appropriate for the school's available resources and technical capacity [4].

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The use of physical records also presents challenges in terms of accessibility and preservation. Important documents stored in filing cabinets or folders may be difficult to retrieve when needed, particularly when staff members must search through numerous records. Physical documents may also be exposed to risks such as accidental damage, loss, deterioration, or unauthorized access. Similarly, information stored in separate spreadsheets can become difficult to maintain when multiple versions of a file exist. Changes made to one file may not automatically appear in another, potentially resulting in inconsistent or outdated information. A centralized database can help reduce these problems by providing a common source of information that authorized users can access according to their assigned roles.

Report preparation is another area where manual processes can place a considerable workload on school personnel. School administrators and teachers may be required to prepare attendance summaries, student reports, academic records, inventory reports, and other documents. When information must be collected from different sources and manually transferred or computed, report preparation can take significant time. Repeated encoding and computation can also increase the possibility of errors. Automating report generation can help reduce repetitive work by allowing information already stored in the system to be organized into reports based on predefined requirements. This can allow school personnel to spend more time on essential educational and administrative responsibilities rather than repetitive data-processing activities.

The growing emphasis on digital transformation in education further highlights the importance of appropriate information systems. The information can be stored in a centralized database. This approach can improve information Philippine education sector has increasingly recognized the role of technology in improving educational services, information management, and access to digital resources. Initiatives such as the Department of Education's MATATAG Agenda and policies supporting digital education emphasize the importance of using technology to improve the delivery and management of education [4]. However, the availability of commercial school management systems does not automatically address the needs of every educational institution. Some existing systems may involve subscription costs, extensive features, complicated configurations, or infrastructure requirements that may not be suitable for smaller private schools. Therefore, there is a need to consider solutions that are affordable, practical, user-friendly, and aligned with the actual operational requirements of the institution [3].

For a school such as TCCIS, a customized web-based school management system can provide an opportunity to address several of these challenges within a single platform. A web-based system can allow authorized users to access school information through computers or other compatible devices using a web browser and an internet connection. Instead of maintaining separate paper records and spreadsheets for different activities, relevant organization and

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make it easier for authorized users to retrieve and update records according to their responsibilities.

The proposed TCCIS School Management System (TCCIS SMS) is designed to integrate several important school management functions into one web-based platform. These functions include student record management, attendance monitoring, inventory management, parent–teacher communication, and automated report generation. By integrating these functions, the system aims to minimize the dependence on scattered paper records and disconnected files. It also seeks to provide a more systematic method of managing information throughout the school's daily operations.

The student record component of the proposed system will provide a centralized location for storing and managing relevant student information. Authorized administrators and teachers can use the system to access information according to their assigned roles. This can reduce the need to search through physical folders and multiple spreadsheets when information is required. A centralized database can also help maintain consistency by allowing updates to be reflected in a common information repository.

The attendance component will support teachers in recording and managing daily student attendance. Instead of relying exclusively on handwritten attendance sheets, teachers can record attendance through the system, allowing attendance information to be stored electronically. This can make attendance records easier to review and organize while providing administrators with access to relevant information for monitoring and reporting purposes.

The inventory component will assist school personnel in monitoring available resources and school materials. It can provide a structured method for recording items, quantities, borrowed materials, returned items, and other relevant inventory information. By maintaining updated inventory records, the school can more easily monitor its resources and identify materials that may require replenishment or further monitoring.

The parent–teacher communication component will provide a centralized means of sharing school-related information. Teachers and authorized school personnel can use the platform to communicate announcements and relevant updates, while parents can access information concerning their children's school activities and academic progress. This component is intended to support more organized and timely communication and reduce dependence on scattered communication channels.

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The automated reporting component will help reduce the time required to prepare common school reports. Information stored within the system can be organized and presented according to the required reporting format. This can minimize repeated encoding and manual computation while helping school personnel produce reports more efficiently. Although the system is not intended to eliminate the need for human review, automation can assist users by reducing repetitive administrative tasks.

Security and controlled access are also important considerations in the development of the proposed system. School information may contain sensitive student, parent, teacher, and administrative data. Therefore, the system will implement role-based access so that users can access functions and information appropriate to their responsibilities. Administrators, teachers, and parents may have different levels of access depending on their roles within the school. This approach can help prevent unauthorized access and support the confidentiality and integrity of school records.

Despite the benefits of implementing a web-based system, certain limitations must also be considered. The effectiveness of the system depends on the availability of suitable hardware, reliable internet connectivity, and users who are capable of operating the system. Technical problems such as internet interruptions, server downtime, power outages, or hardware problems may temporarily affect system availability. In addition, users who are accustomed to manual procedures may require orientation and training before they can confidently use the new system. These considerations emphasize the importance of designing the system with a simple, understandable, and user-friendly interface.

The development of TCCIS SMS therefore focuses not only on digitizing existing processes but also on creating a practical solution that reflects the actual needs and capabilities of the school. The system is intended to support administrators, teachers, parents, and other authorized personnel by providing appropriate access to information and functions. Through the integration of student records, attendance, inventory, communication, and reporting, the proposed system aims to create a more organized and efficient approach to school management.

Ultimately, this project responds to the need for a practical and accessible digital solution for Timothy's Cradle Christian International School Inc. By replacing or reducing dependence on manual records and disconnected files, TCCIS SMS aims to improve the efficiency of daily school operations, enhance the accuracy and accessibility of information, strengthen communication between the school and parents, and support better management of school resources. The proposed system also provides a foundation for the school's gradual digital transformation by introducing technology that is specifically designed around its operational requirements. Through this project, the researchers seek to demonstrate how a properly designed web-based school management system can support the institution's administrative

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and academic activities while providing a more organized, reliable, and sustainable approach to managing school information [5].

1.2 Statement of the Problem

Timothy's Cradle Christian International School Inc. currently relies on a combination of manual records, physical documents, logbooks, and basic spreadsheets to manage various administrative and academic activities. These traditional methods are used in several important school processes, including student record management, attendance monitoring, inventory tracking, communication with parents and teachers, and preparation of administrative and academic reports. Although these approaches provide a familiar and accessible way of handling school information, they require considerable time and effort from teachers, administrators, and other school personnel. As the school continues to manage increasing amounts of information, the limitations of these manual and fragmented processes become more evident.

The use of paper-based records and separate digital files can make it difficult to maintain information in an organized and consistent manner. Important records may be stored in different locations, such as filing cabinets, logbooks, folders, and individual spreadsheet files. As a result, school personnel may need to search through several sources before finding the information they need. Updating records can also require repeated encoding, particularly when the same information is needed for different purposes. This may increase the possibility of duplicate entries, incomplete information, inconsistent records, and human errors.

Manual processes can also affect the timeliness of school operations. Attendance information, student records, inventory updates, and other important data may not always be immediately available when needed. Teachers and administrators may need to manually consolidate information before it can be reviewed or used for decision-making. This can delay the preparation of reports and make it more difficult to monitor important concerns, such as student attendance, available school resources, and academic information.

Another concern is the vulnerability of physical and fragmented records. Paper documents may be misplaced, damaged, or deteriorated over time, while spreadsheet files may be accidentally modified, duplicated, or stored in different versions. Without a centralized information management system, maintaining accurate and updated records becomes more challenging. The absence of a single source of information may also make it difficult to determine which record contains the most current data.

Communication is another area affected by fragmented processes. Important announcements, student updates, and other school-related information may be communicated through different channels. When there is no centralized platform for school-to-parent and teacher-to-parent communication, messages may be missed, delayed, or difficult to retrieve. This can affect the ability of parents and teachers to remain consistently informed about matters concerning students and school activities.

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In addition, preparing school reports through manual data consolidation can increase the workload of school personnel. Information may need to be gathered from different documents and spreadsheets, checked for accuracy, manually computed, and transferred into report formats. Repetitive encoding and calculations can consume valuable time and increase the possibility of errors. When corrections are required, personnel may also need to review several records and update multiple files, further increasing the amount of work involved.

These challenges indicate the need for a more centralized, organized, and efficient approach to managing the school's information and daily operations. A web-based school management system can provide a unified platform where authorized users can record, update, retrieve, and manage information according to their respective roles. By integrating student records, attendance monitoring, inventory management, parent–teacher communication, and report generation, the proposed system can reduce dependence on scattered manual processes and improve the overall organization of school information.

Therefore, this study proposes the development of the TCCIS School Management System (TCCIS SMS), a web-based system designed to address the identified challenges in the current school management processes of Timothy's Cradle Christian International School Inc. The system aims to provide a centralized platform that can support more efficient record management, timely attendance monitoring, systematic inventory tracking, organized communication, and automated report generation. Through the proposed system, the researchers seek to contribute to improved efficiency, accuracy, accessibility, consistency, and management of school information while providing a practical digital solution suited to the needs of the institution.

The school's reliance on manual and fragmented processes presents five critical challenges:

1.The school's reliance on manual and fragmented processes presents five critical challenges:

Attendance recording is slow, prone to human error, and compiled late, delaying timely intervention for absenteeism or chronic tardiness [1].

2.Student records are scattered across physical files and disconnected spreadsheets, making retrieval difficult, data integrity hard to verify, and information vulnerable to loss or damage [2].

3.Inventory and asset monitoring lack systematic tracking mechanisms, leading to unrecorded movement, unexpected shortages, unnecessary overstocking, and poor accountability during audits [3].

4.Communication between the school and families relies on informal, unstructured channels — resulting in missed announcements, delayed feedback, and limited parental engagement [5].

5.Report generation requires repeated manual data consolidation and computation, consuming significant staff time and producing inconsistent, error-prone outputs [4].

1.3 Objectives of the Study

General Objective

The general objective of this study is to design, develop, implement, and evaluate a web-based integrated school management system for Timothy's Cradle Christian International School Inc. (TCCIS) that will help improve the efficiency, accuracy, accessibility, organization, and management of the school's administrative and academic operations. The proposed system is intended to address the limitations associated with the school's existing manual, paper-based, and fragmented processes by providing a centralized digital platform for managing essential school information and activities.

The proposed TCCIS School Management System (TCCIS SMS) will integrate important school management functions into a single web-based platform. These functions will include student record management, attendance monitoring, inventory and asset tracking, parent-teacher communication, and automated report generation. Through this integration, the system aims to reduce the need for maintaining separate physical documents, logbooks, and disconnected spreadsheet files. It will provide authorized users with a more organized method of recording, updating, retrieving, and managing information required for daily school operations.

The system will also aim to improve the accessibility of school information by providing authorized administrators, teachers, and parents with role-based access to relevant features and records. Administrators will be able to manage and monitor important school information, teachers will be able to perform academic and attendance-related tasks, and parents will be able to access appropriate information concerning their children and school activities. The implementation of role-based access is intended to ensure that users can interact with the system according to their responsibilities while helping protect school information from unauthorized access.

In addition, the proposed system aims to automate selected repetitive administrative tasks. Attendance information can be recorded and organized digitally, student information can be maintained within a centralized database, inventory records can be updated systematically, and reports can be generated using information stored in the system. By reducing repetitive manual encoding and computation, the system seeks to lessen the administrative workload of school personnel and allow them to allocate more time to other important academic and administrative responsibilities.

The system also aims to improve the accuracy and consistency of school information. A centralized database can help minimize duplicate records and reduce inconsistencies that may occur when information is maintained across multiple documents and files. Authorized users will be able to update information within the appropriate system modules, allowing relevant records to be maintained in a more systematic manner. The system will also provide mechanisms for

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organizing information so that records can be retrieved more efficiently when needed for monitoring, decision-making, or reporting.

Another objective is to strengthen communication between the school and parents by providing a centralized communication feature. The system will support the sharing of school announcements, relevant student information, and other appropriate updates between authorized school personnel and parents. This feature is intended to reduce reliance on scattered or informal communication channels and provide a more organized way of delivering school-related information.

The proposed system will also support more effective monitoring of school resources through its inventory management function. The system will provide a structured method for recording school supplies, equipment, learning materials, and other assets. It will assist authorized personnel in monitoring item quantities, movements, borrowing, returns, and other relevant inventory information. This can help the school maintain more accurate inventory records and improve accountability for its available resources.

Furthermore, the study aims to evaluate the developed system to determine whether it meets the identified requirements and operational needs of TCCIS. The evaluation will consider relevant characteristics such as functionality, usability, efficiency, reliability, accessibility, and overall suitability for the intended users. Feedback from the system's intended stakeholders may be considered in identifying areas that require improvement and refinement. This evaluation will help determine whether the developed system can effectively support the school's existing processes and whether it provides a practical alternative to selected manual procedures.

Overall, the study seeks to develop a centralized, integrated, user-friendly, and practical web-based school management system that responds to the specific needs of Timothy's Cradle Christian International School Inc. By integrating essential administrative and academic functions into one platform, the proposed TCCIS SMS aims to improve the management of school information, reduce repetitive manual work, facilitate communication, support more efficient reporting, strengthen record organization, and provide a foundation for the school's continued adoption of digital technologies [1].

Specific Objectives:

1. Automate attendance recording and monitoring through real-time data capture, percentage calculation, and automated alerts for absenteeism or irregular attendance [9].

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2. Centralize and secure all student records — including personal profiles, enrollment history, grades, and attendance — in a searchable, role-restricted digital database [2].
3. Implement inventory tracking with transaction logging, real-time stock monitoring, automated low-stock notifications, and complete asset lifecycle management [13].
4. Establish a secure, centralized communication portal for announcements, direct messaging, and student progress updates between parents and teachers [8].
5. Enable automated, customizable report generation for attendance summaries, grade sheets, student master lists, inventory status, and compliance documents in standard export formats [5].

1.4 Scope and Delimitation

This study focuses on the design, development, implementation, and evaluation of a web-based integrated school management system for Timothy's Cradle Christian International School Inc. (TCCIS). The proposed system is developed to address the identified challenges associated with the school's existing manual, paper-based, and fragmented processes in managing student information, attendance, academic records, inventory, communication, and report preparation. The system aims to provide a centralized digital platform where authorized users can efficiently record, manage, retrieve, update, and monitor school-related information. By integrating these functions into a single system, the project seeks to reduce repetitive manual work, improve the organization and accessibility of information, and support more efficient administrative and academic operations.

The system covers student enrollment and profile management, which allows authorized personnel to create, update, view, and manage student records. Relevant information may include basic student details, enrollment information, parent or guardian information, and other school-related data required for maintaining student profiles. Instead of relying primarily on physical folders and separate spreadsheets, the proposed system provides a centralized database where authorized users can access current student information. This feature is intended to make record retrieval more efficient and reduce the possibility of duplicate, incomplete, or inconsistent records.

The system also includes daily attendance logging through a web-based interface. Teachers or other authorized personnel can record student attendance according to the appropriate date and attendance status. Attendance information will be stored electronically and organized within the system, allowing authorized users to review attendance history and generate relevant attendance information when required. This feature is intended to reduce the time spent on manual attendance consolidation and provide a more organized method of monitoring student attendance, absenteeism, and tardiness.

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Another function included within the scope of the system is grade encoding and viewing. Authorized teachers can enter and manage student grades using the web-based platform. Appropriate users can view academic information based on their assigned roles and permissions. The system is intended to provide a centralized method of maintaining academic records and reduce the need to maintain separate grade sheets or disconnected spreadsheet files. However, the grade-related functions are limited to recording, managing, viewing, and reporting academic information and do not constitute a complete online learning or teaching platform.

The system also covers inventory item and transaction management. Authorized school personnel can record school supplies, equipment, learning materials, and other resources that require monitoring. Inventory transactions may include the addition of new items, issuance, borrowing, returning, and updating of quantities or item information. The system will maintain digital inventory records to help authorized personnel determine available quantities and monitor the movement of school resources. This feature aims to improve accountability, reduce the possibility of unrecorded inventory movement, and support more systematic resource management.

For communication purposes, the proposed system includes parent–teacher direct messaging and announcement publishing. Authorized teachers and parents can communicate through the system's messaging functionality, while administrators or authorized school personnel can publish school announcements and relevant updates. This feature is intended to provide a more organized and centralized communication channel for school-related information. It aims to reduce dependence on informal or scattered communication methods and help parents remain informed about announcements and matters concerning their children.

The system also includes exportable reporting functionality. Authorized users can generate and export reports based on information stored within the system. Depending on the user's role and permissions, available reports may include student information, attendance, academic records, inventory information, and other reports supported by the system. The reporting feature is intended to reduce repetitive manual data consolidation and computation by using information already stored in the centralized database. However, generated reports may still require review and verification by authorized school personnel before being used for official purposes.

To protect school information and control access to system functions, the proposed system implements role-based access control. Administrators, teachers, parents, and other authorized personnel will have different permissions depending on their responsibilities within the school. Administrators may have access to broader management functions, teachers may access student, attendance, grade, and communication features relevant to their responsibilities, while parents may be limited to information concerning their children and appropriate school

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announcements. This access structure is intended to help protect confidential information and reduce unauthorized access to records [3].

The proposed system is designed to be accessible through standard web browsers on desktop computers, laptops, tablets, and compatible mobile devices. Users will not be required to install a separate desktop application to access the web-based platform. The system may support either cloud-based deployment or local-area-network (LAN) deployment, depending on the school's available infrastructure, hosting environment, network configuration, and technical requirements [2]. Cloud-based deployment may allow users to access the system through an internet connection, while LAN deployment may allow access within the school's local network.

Although the system is designed to provide several integrated school management functions, the study is delimited to the specific features identified in the project requirements. It does not attempt to develop a complete enterprise-level school management platform containing every possible administrative, financial, academic, or learning-related function. The project focuses only on the core functions required to address the identified problems of TCCIS. Additional features that are not identified as part of the project's requirements are excluded from the current version.

The proposed system does not include a full Learning Management System (LMS). It is not designed to deliver complete online classes, host online lectures, conduct live virtual classroom sessions, manage comprehensive digital learning materials, or handle the submission and assessment of digital assignments [4]. While the system includes grade management and selected academic functions, its primary purpose is school information and administrative management rather than complete online education delivery.

The system also does not include online payment gateways or real-time financial transaction processing. Parents or other users will not be able to make tuition payments, transfer funds, process credit or debit card payments, or conduct other financial transactions through the system. If fee-related information is included, it will be limited to record-keeping and monitoring purposes and will not function as a complete accounting, billing, payroll, or financial management system [3].

Another delimitation concerns offline functionality. The current version of the system requires a stable internet connection when deployed through a cloud-based environment. Users must be connected to the internet to log in, access records, submit information, communicate with other users, and perform system operations. The system does not currently provide an offline data-capture feature that allows users to continue entering information without an internet connection and automatically synchronize the data after connectivity is restored [10].

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The operation of the system is also dependent on the availability of appropriate hardware, electricity, network infrastructure, and internet services. Technical problems involving computer or mobile device failures, server problems, damaged network equipment, power interruptions, router or connectivity problems, and Internet Service Provider (ISP) disruptions are outside the direct scope of the study [1]. The project assumes that users have access to functional devices, appropriate network infrastructure, and the required power and connectivity needed to operate the system.

The study also does not include responsibility for maintaining or repairing the school's physical hardware infrastructure. The researchers' primary responsibility is focused on the design, development, implementation, and evaluation of the software system. Hardware replacement, electrical repairs, network installation, internet subscription, server maintenance outside the defined system environment, and ISP-related troubleshooting are considered external operational concerns.

Furthermore, the system is not intended to replace the roles and responsibilities of teachers, administrators, parents, or other school personnel. The system serves as a technological tool to support users in managing information and performing routine tasks. Human judgment, verification, approval, and decision-making remain necessary. For example, teachers and administrators are still responsible for verifying the correctness of attendance and grade information entered into the system, while authorized personnel remain responsible for checking inventory records and validating reports before official use.

The system's effectiveness is also dependent on the accuracy and completeness of information entered by authorized users. A centralized database can help organize and manage information, but incorrect information entered into the system may still result in inaccurate records or reports. Therefore, proper user orientation, responsible data entry, regular updating, and verification of information remain important factors in ensuring reliable system operation.

In addition, advanced technologies and integrations are outside the current delimitation of the study. The system does not include biometric attendance devices, artificial intelligence-based student assessment, facial recognition, automated SMS services, advanced predictive analytics, comprehensive accounting functions, payroll management, mobile applications developed separately from the web platform, or direct integration with external government databases unless such features are specifically identified as requirements of the current project.

The study is therefore limited to developing a web-based integrated school management system specifically designed for the identified operational needs of Timothy's Cradle Christian

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International School Inc. Its primary focus is on student enrollment and profile management, attendance logging, grade encoding and viewing, inventory management, parent–teacher communication, announcement publishing, role-based access control, and exportable reporting. These boundaries allow the researchers to concentrate on the major challenges identified in the existing school processes while keeping the proposed system practical, manageable, and aligned with the available resources and requirements of the institution.

Overall, the scope and delimitation establish the specific functions, users, technologies, and operational conditions covered by the study. The proposed TCCIS School Management System is intended to provide a centralized and user-friendly platform that supports essential school management activities while recognizing that certain functions, technical requirements, and external services remain outside the project's coverage. These defined boundaries provide a clear basis for the development and evaluation of the system and ensure that the project remains focused on addressing the most relevant information-management challenges of the school.

1.5 Significance of the Study

The development of the TCCIS School Management System (TCCIS SMS) is significant because it aims to provide a centralized and integrated digital platform that can improve the management of academic and administrative information at Timothy's Cradle Christian International School Inc. The proposed system addresses several challenges associated with manual records, paper-based processes, disconnected spreadsheets, and informal communication channels. By integrating student records, attendance monitoring, grade management, inventory management, communication, and reporting into a single web-based platform, the system can provide benefits to various stakeholders of the school.

Administrators

The proposed system can provide school administrators with a more organized and centralized method of managing important school information and monitoring daily operations. Instead of relying on multiple physical documents, logbooks, and separate spreadsheet files, administrators can access relevant information through a centralized database. This can make it easier to monitor student records, attendance, academic information, inventory, announcements, and reports.

The system can also support administrators in generating standardized and organized reports based on information stored in the system. Automated reporting can reduce the time spent collecting and consolidating information from different sources. It can also support

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administrative activities such as monitoring, auditing, documentation, and compliance-related preparation. By having access to updated information, administrators can make more informed decisions based on available school records and operational data [4].

Furthermore, role-based access can help administrators manage user permissions and protect school information. The system can provide administrators with greater visibility over relevant activities while limiting access to authorized users. This can contribute to improved data governance, accountability, and organization of school records.

Teachers

Teachers can benefit from the system by reducing the amount of time and effort required to perform repetitive administrative tasks. Through the web-based attendance module, teachers can record and manage student attendance electronically rather than relying entirely on handwritten attendance sheets. Attendance records can then be stored and organized for future review and reporting.

The grade management feature can also assist teachers in encoding and viewing students' academic information. Instead of maintaining separate grade files or repeatedly transferring information between documents, teachers can manage relevant records through the centralized system. This can help reduce repetitive data entry and support more organized academic record management.

The reporting functionality can further assist teachers by reducing the time required to manually consolidate information for reports. By allowing teachers to retrieve relevant student, attendance, and academic information from one platform, the system may reduce administrative workload and allow teachers to devote more time to instruction, lesson preparation, classroom activities, assessment, and student support [5].

In addition, the communication features can provide teachers with a more organized channel for communicating with parents. Teachers can send relevant messages and updates through the system, helping facilitate communication regarding student progress, attendance, announcements, and other school-related concerns.

Parents

Parents can benefit from the proposed system by gaining more convenient and organized access to information concerning their children. Depending on their assigned permissions,

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parents can view relevant student information, attendance records, academic grades, school announcements, and other information made available by authorized school personnel.

The communication feature can provide parents with a direct channel for communicating with teachers or authorized school personnel. This can reduce dependence on scattered communication methods and make it easier for parents to receive important updates and provide appropriate feedback.

Access to timely school information can also encourage greater parental involvement in a child's education. By being able to monitor attendance, academic performance, announcements, and other relevant information, parents can become more aware of their child's school activities and respond to concerns when necessary [8]. The system therefore has the potential to strengthen communication and cooperation between parents and the school while supporting a more active home-school partnership.

Students

Students can benefit indirectly from the proposed system through improved management of their academic and school-related information. Although students may not have the same level of access as administrators or teachers, the centralized management of student records can help ensure that relevant academic and attendance information is properly organized and maintained.

Accurate and accessible records can support the timely monitoring of students' attendance and academic performance. Better-organized information can also help teachers and administrators identify records that require attention and provide appropriate support when necessary. In addition, improved communication between parents and teachers can contribute to a more coordinated support system for students.

The inventory management component may also benefit students by helping the school maintain better records of learning materials, classroom resources, and other educational resources. More systematic resource management can support the availability and accountability of materials needed for school activities. Overall, the proposed system can contribute to a more organized, responsive, and reliable educational environment for students [3].

Future Researchers

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The study can serve as a useful reference for future researchers who intend to develop similar information systems for schools, particularly small private educational institutions with limited financial, technical, and human resources. The development process, system requirements, database structure, system modules, documentation, testing procedures, and evaluation results can provide a foundation for future studies involving school management systems.

Future researchers may use the study as a reference when identifying common challenges associated with manual school management processes and determining appropriate digital solutions. They may also build upon the proposed system by introducing additional functions or technologies, such as mobile applications, biometric attendance, advanced analytics, automated notifications, online payment integration, or learning management features.

The study can also contribute to future research on the use of affordable and customized web-based technologies in small educational institutions. By documenting the development and evaluation of a school management system designed around the specific needs of TCCIS, the research can provide a framework that future researchers may adapt to other schools with similar operational challenges [1].

Overall Significance

Overall, the proposed TCCIS School Management System is significant because it provides an integrated approach to managing essential school information and operations. For administrators, it can support centralized management, monitoring, reporting, and data governance. For teachers, it can reduce repetitive administrative work and provide more organized tools for managing attendance, grades, and communication. For parents, it can improve access to relevant student information and provide a more organized communication channel with the school. For students, it can contribute to more accurate record management, improved communication, and better resource organization. For future researchers, it can provide documented information and a development framework that may serve as a reference for similar projects.

Through the integration of these functions into one web-based platform, the study aims to demonstrate how an appropriately designed information system can support the digital transformation of a small educational institution while remaining focused on its specific operational needs and available resources.

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CHAPTER II

REVIEW OF RELATED LITERATURE AND STUDIES

2.1 School Management & Student Information Systems

Integrated School Management Systems (SMS) and Student Information Systems (SIS) have become increasingly important tools for educational institutions seeking to improve the efficiency, accuracy, and accessibility of their administrative and academic operations. These systems are designed to bring different school processes into a centralized digital environment, allowing information to be collected, stored, processed, updated, and retrieved through a unified platform. Instead of depending on separate paper records, logbooks, spreadsheets, and independent databases, an integrated system allows related information to be connected and managed through a common database structure.

Integrated school management platforms can consolidate essential functions such as student enrollment, student profile management, grading, class scheduling, attendance monitoring, reporting, inventory management, communication, and compliance-related activities into a unified digital ecosystem. By connecting these functions, schools can reduce the occurrence of data silos, where information is stored separately and cannot be easily accessed or shared between departments. Data silos can result in repeated data entry, inconsistent records, outdated information, and additional administrative work. A centralized system addresses these issues by allowing authorized users to work with information maintained within a common database [6].

The integration of school functions can also reduce data duplication and repetitive encoding. In a manual or fragmented environment, the same student information may have to be entered into multiple documents or files. For example, a student's personal information may be recorded during enrollment and then entered again into attendance records, grading records, class lists, and reports. Repeated encoding increases the possibility of typographical errors and inconsistencies. In an integrated system, information can be entered once and associated with other relevant modules through unique student identifiers. This allows information to be reused across appropriate functions while reducing unnecessary duplication [6].

Another important benefit is the creation of a single reliable source of information. When school records are distributed across different files and physical documents, it may be difficult for personnel to determine which version contains the most current information. A centralized database can provide a common source from which authorized users retrieve information.

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When properly maintained, this can improve consistency across student records, attendance data, grades, inventory information, and reports. It can also make information retrieval faster because users do not need to search through several separate sources.

Integrated systems can also support better administrative decision-making. School administrators often need current information when monitoring student attendance, academic performance, enrollment, resources, and other operational activities. When information is available in an organized digital system, administrators can retrieve relevant records more efficiently and use them when preparing reports or making administrative decisions. Teachers can similarly access information relevant to their responsibilities without relying entirely on paper documents or manually maintained files.

Cloud-based architectures provide additional advantages for institutions adopting school management systems. A cloud-based system stores application data and services on remotely managed infrastructure, allowing authorized users to access the system through an internet connection. This approach can reduce the need for schools to purchase and maintain extensive on-premise servers and other infrastructure. It can also allow users to access the system from different locations and devices, provided that appropriate authorization and connectivity are available [3].

For small educational institutions, cost-efficiency and scalability are particularly important considerations. Maintaining a large on-premise information technology infrastructure may require significant investment in servers, networking equipment, software, maintenance, and technical personnel. Cloud-based solutions can potentially reduce some of these infrastructure requirements and allow schools to adopt digital systems according to their available resources [3]. As the number of students, teachers, and records increases, a scalable system can also be adjusted to accommodate additional users and information without requiring the institution to completely replace its existing system.

However, the implementation of cloud-based systems also requires consideration of internet connectivity, data security, privacy, user authentication, and system availability. Schools must ensure that only authorized individuals can access sensitive information and that appropriate measures are in place to protect stored data. Reliable connectivity is also important because users may be unable to access cloud-based services during internet interruptions. Therefore, the advantages of cloud-based school management systems must be considered together with the institution's technical infrastructure and operational requirements.

In the Philippine educational context, the importance of effective data management is also reflected in discussions surrounding education reform and digital transformation. EDCOM II has emphasized the need for better education data systems and more interoperable approaches to

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managing information within the Philippine education sector [1]. Interoperable and standardized data frameworks can help reduce administrative burdens by allowing information to be organized consistently and used across different processes. Standardization can also improve data quality by establishing common formats, definitions, identifiers, and procedures for collecting and managing information.

The importance of interoperability is particularly relevant because schools generate information through many different activities. Enrollment creates student information, attendance systems generate daily attendance records, teachers produce grades and academic information, and administrative personnel prepare reports based on these records. If these processes operate independently, school personnel may need to repeatedly transfer information between systems or documents. This can increase workload and create opportunities for errors. An integrated and interoperable system can allow related information to move more efficiently between modules while maintaining consistent student identifiers and data structures [6].

Standardized data management can also support more reliable reporting. When information is collected and stored using consistent formats, it becomes easier to retrieve, summarize, and generate reports. This is particularly important when schools are required to prepare administrative reports or provide information to authorized educational authorities. A structured system can reduce the need for repeated manual consolidation and help ensure that reports are based on consistent information [1].

Furthermore, an integrated school management system can improve administrative efficiency by automating repetitive tasks. Attendance information can be stored automatically after being recorded, student information can be retrieved when preparing reports, and inventory records can be updated through defined transactions. Automated processes do not eliminate the need for human supervision, but they can reduce repetitive activities and allow teachers and administrative personnel to focus more on tasks that require professional judgment and direct interaction with students and families.

The integration of school management functions can also improve information accessibility. Authorized users can retrieve records through a web interface instead of physically searching through filing cabinets or multiple folders. This can be particularly useful when information is needed immediately for monitoring, reporting, or communication. At the same time, accessibility must be balanced with security through authentication and role-based permissions to ensure that sensitive information is only available to appropriate users.

Overall, the literature indicates that School Management Systems and Student Information Systems can provide significant organizational benefits by integrating administrative and academic functions into a centralized digital environment. The use of centralized databases can

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reduce data duplication and fragmentation, while cloud-based architectures can improve accessibility, scalability, and potentially reduce infrastructure requirements [3][6]. In the Philippine context, the emphasis on interoperable and standardized education data further demonstrates the importance of developing systems that can manage information consistently and efficiently [1].

These concepts are relevant to the development of the TCCIS School Management System (TCCIS SMS) for Timothy's Cradle Christian International School Inc. The proposed system applies the principles of centralized data management, system integration, standardized information handling, and role-based access to selected school processes. By connecting student records, attendance, grades, inventory, communication, and reporting within one web-based platform, the system aims to reduce fragmentation and repetitive manual work while providing authorized users with more organized and accessible school information.

The iSercan (2019) web-based student information system demonstrates how centralized digital platforms can improve the management and accessibility of student-related information within educational institutions. By integrating important academic and administrative data such as grades, class schedules, attendance records, and financial information into a single web-based portal, the system reduces the need for schools to depend heavily on paper-based documents and disconnected records. Centralizing these functions allows authorized users to retrieve information more quickly and efficiently, minimizing the time spent searching through physical files or multiple data sources. It also promotes greater transparency between the school and families by allowing relevant information to be accessed through a structured and organized platform [2].

The study further emphasizes the importance of role-based access in managing information within a school environment. Different users, such as administrators, teachers, students, and parents, may require access to different types of information and system functions. Role-based permissions help ensure that users can only view or manage information appropriate to their responsibilities, thereby reducing the possibility of unauthorized access or accidental modification of sensitive records. This approach is particularly important in student information systems because academic, personal, attendance, and financial information must be handled with appropriate levels of confidentiality and security [2].

Another significant aspect highlighted by iSercan (2019) is the importance of usability and accessibility. A system may provide extensive functionality, but it can be difficult to adopt if its interface is complicated or unfamiliar to users. Structured and user-friendly interfaces can help users navigate the system more easily, particularly those with limited digital literacy or experience with information systems. Clear menus, organized information, and straightforward workflows can reduce confusion and minimize errors during data entry and retrieval. Automated notifications can also improve communication by informing users about relevant updates, such

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as changes in schedules, attendance information, academic records, or other school-related activities [2].

Security is also an essential consideration in the implementation of a web-based student information system. Since the platform handles sensitive student and school information, appropriate security mechanisms are necessary to protect data from unauthorized access, modification, or disclosure. Encryption can help protect information during storage or transmission, while authentication and access-control mechanisms restrict system functions to authorized users. These measures contribute to maintaining the confidentiality, integrity, and reliability of school records [2].

The findings of iSercan (2019) are relevant to the development of the proposed TCCIS School Management System because they demonstrate the benefits of combining centralized information management, role-based access, user-friendly interfaces, automated notifications, and security controls within a single web-based platform. Similar principles can be applied to TCCIS by integrating student records, attendance, grades, inventory information, parent–teacher communication, and reporting functions into one system. Through this approach, the proposed system can reduce fragmented record-keeping, improve information retrieval, support more efficient school operations, and provide authorized stakeholders with timely and secure access to relevant information.

Chen and Huang (2024) emphasize that true system integration extends beyond the simple exchange or sharing of information between different modules. Effective integration requires a unified data structure in which information is organized according to consistent formats, definitions, and relationships. The use of common identifiers is particularly important because it allows the system to recognize the same student, teacher, class, or transaction across different functions. With a unified data model, information entered into one module can be reliably connected to other modules without requiring users to repeatedly encode the same information [6].

For example, when a student is enrolled and their information is recorded in the enrollment module, the same student record should automatically become available to the appropriate attendance, grading, reporting, and other authorized functions. Teachers should not need to create a separate student record every time they record attendance or grades. Instead, the modules should communicate through a shared database and consistent identifiers. This reduces duplicate data entry and helps ensure that changes made to a student's information are reflected throughout the system. Such integration also improves data consistency because users are working with the same source of information rather than maintaining separate copies of records [6].

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Seamless communication between modules can also improve the efficiency of administrative and academic processes. When information flows automatically between connected functions, routine tasks such as updating student profiles, preparing attendance summaries, generating academic reports, and maintaining records can be completed with less manual intervention. This reduces repetitive work and allows school personnel to focus more on tasks that require human judgment and interaction. Furthermore, centralized and interconnected information makes it easier for authorized users to retrieve updated records and generate reports based on consistent data [6].

In contrast, the absence of interoperability can create isolated departments or functional areas in which each module maintains its own records. This can result in repeated data entry, inconsistent information, and difficulties in determining which record is the most current. For instance, if a student's information is changed in an enrollment record but is not reflected in the attendance or grading records, discrepancies may occur. Re-entering the same information in multiple locations also increases the possibility of typographical and encoding errors. As these errors are transferred to other processes, they can affect reports and other school operations, creating a chain of inaccurate or outdated information [6].

The concept presented by Chen and Huang (2024) is particularly relevant to the proposed TCCIS School Management System. The system is designed not merely as a collection of separate features but as an integrated platform in which student records, enrollment, attendance, grades, inventory, communication, and reporting functions are connected through a centralized database. By applying unified data structures and common identifiers, the proposed system can minimize duplicate encoding and improve consistency across its modules. This approach supports the goal of creating a more efficient and reliable school management environment in which authorized users can access updated information without maintaining multiple independent records [6].

For academic advising, a web-based expert system developed at the Technological Institute of the Philippines reduced manual curriculum evaluation from days to near-instant results by encoding institutional rules directly into system logic — ensuring consistency, fairness, and timeliness regardless of staff availability or turnover [7]. This underscores the value of embedding school-specific policies and workflows directly into custom-built management systems [7].

2.2 Attendance Monitoring Systems

Attendance monitoring is an essential component of school management because it provides administrators and teachers with information about student participation, punctuality, and regularity in attending classes. Traditional attendance practices, such as manual roll calls, paper

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attendance sheets, and logbooks, require teachers to spend considerable time recording and organizing attendance information. These methods can become increasingly difficult to manage as the number of students and classes increases. Manual recording is also susceptible to human error, including incorrect marking, incomplete entries, duplicate records, and inaccurate calculations of attendance totals. In addition, paper-based attendance records can be misplaced, damaged, or difficult to retrieve when historical information is needed. Another concern is proxy attendance, in which one student records or represents the attendance of another student, reducing the reliability of the attendance data [9].

Digital attendance monitoring systems have been developed to address many of these limitations by automating the process of recording and storing attendance information. Among the available approaches, QR-code-based attendance systems provide a relatively low-cost and accessible alternative for educational institutions. In this approach, a QR code can be generated for a particular class or attendance session and scanned using a mobile device. The student's attendance information can then be validated and recorded in a centralized database. Compared with manual roll calls, this process can reduce the amount of time required for attendance recording and minimize repetitive encoding by teachers or administrative personnel. Centralized storage also allows authorized users to retrieve attendance records more efficiently for monitoring and reporting purposes [9].

The effectiveness of QR-based attendance systems can be further improved through security mechanisms that control when and how attendance codes can be used. Time-bound QR codes, for example, can be configured to remain valid only during a specified attendance period. Once the designated attendance window has closed, the code can automatically expire and become invalid. Encrypted or dynamically generated codes can provide additional protection against unauthorized reuse or sharing. These mechanisms can help reduce delayed attendance submissions and other forms of misuse while maintaining a relatively simple process for teachers and students [9].

However, the implementation of digital attendance systems must also consider connectivity limitations. In communities where internet access may be intermittent or unreliable, systems that depend entirely on continuous online connectivity may interrupt attendance recording. Hybrid offline–online architectures provide an alternative by allowing attendance information to be temporarily stored on a local device or local network when an internet connection is unavailable. Once connectivity is restored, the stored records can be synchronized with the central database. This approach helps maintain continuity in attendance operations and reduces the possibility of losing records because of temporary network interruptions [10].

Additional verification mechanisms can strengthen the reliability of attendance information. Geolocation verification, for example, can be used to determine whether an attendance transaction occurs within an authorized geographical area, such as the school premises. This

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can help restrict attendance recording to an appropriate location and provide another layer of validation. Automated notifications can also improve the usefulness of attendance monitoring by informing parents or authorized administrators when a student is recorded as absent or tardy. Timely notifications allow parents and school personnel to become aware of attendance concerns sooner instead of waiting for manually prepared attendance summaries [10].

Biometric attendance systems provide another approach by using unique physiological characteristics for identification. Because biometric characteristics are associated with individual users, these systems can provide a stronger form of identity verification than methods that rely only on manually entered names or shared credentials. However, biometric solutions may require specialized hardware, additional implementation costs, maintenance, privacy considerations, and technical expertise. For smaller educational institutions with limited financial and technical resources, these requirements may make biometric systems more difficult to implement. QR-code-based systems may therefore provide a more accessible alternative when the institution's primary objective is to improve attendance recording without requiring specialized biometric equipment [11].

Modern attendance monitoring systems can also move beyond simply recording whether a student is present or absent. Digitally stored attendance data can be analyzed to identify patterns such as frequent absences, repeated tardiness, attendance percentages, and changes in attendance behavior over time. Automated summaries and trend reports can help teachers and administrators identify students who may require closer monitoring or appropriate intervention. Instead of waiting until attendance problems become significant, school personnel can use accumulated attendance information to recognize patterns earlier and communicate with parents when necessary [12].

Integration with other school management functions can further increase the value of attendance information. When attendance records are connected to student profiles, academic grades, and reporting modules, authorized users can access a more complete view of a student's school participation. For example, attendance data can be included in student reports without requiring teachers to manually transfer information from separate attendance sheets. Integration can also reduce duplicate encoding and improve consistency because attendance information is stored in the same centralized environment as other relevant student records [12].

For the proposed TCCIS School Management System, these concepts demonstrate the importance of developing attendance monitoring as an integrated component rather than as an isolated feature. The system can provide a web-based interface for teachers to record daily attendance and store the resulting information in a centralized student database. Attendance records can then be used for monitoring, report generation, and authorized parent access. Although the proposed system may not initially implement biometric verification or advanced geolocation features, the principles of automated recording, centralized storage, access control,

timely reporting, and future scalability provide a foundation for improving the efficiency and reliability of attendance management at TCCIS.

Overall, the literature indicates that the transition from manual attendance recording to digital monitoring can address several common challenges associated with paper-based processes, including time consumption, repetitive encoding, record retrieval, and data consistency. QR-based approaches, secure validation mechanisms, offline–online synchronization, automated notifications, and attendance analytics represent different strategies that can be considered according to an institution's resources and requirements. For a small school such as TCCIS, an appropriately designed and user-friendly web-based attendance module can provide practical improvements while remaining aligned with the broader goal of developing an integrated school management system [9–12].

2.3 School Inventory & Asset Management

School inventory and asset management is an important part of educational administration because schools depend on different types of resources to support teaching, learning, and daily operations. These resources include consumable supplies such as paper, pens, cleaning materials, and instructional materials, as well as durable assets such as computers, printers, projectors, desks, chairs, books, laboratory equipment, and other school property. Proper inventory management enables school personnel to determine what resources are available, where they are located, who is responsible for them, and when additional supplies or maintenance may be required. Without an organized inventory process, schools may experience difficulties in monitoring resources and maintaining accurate records of their assets [13].

Traditional inventory tracking commonly relies on paper logs, handwritten records, and spreadsheets. Although these methods may be suitable for managing a small number of items, they become increasingly difficult to maintain as the number of students, classrooms, employees, and school resources increases. Personnel may need to repeatedly check physical records, update spreadsheets, and manually compare recorded quantities with actual stock. These activities consume time and may result in inconsistent or outdated information when transactions are not recorded immediately. Errors may also occur when items are incorrectly counted, recorded under the wrong category, or omitted from the inventory records. Such limitations can lead to discrepancies between the actual inventory and the information maintained by the school [13].

Another challenge associated with manual inventory management is the difficulty of tracking the movement of school property. Items may be transferred between classrooms, borrowed by teachers, returned to storage areas, or assigned to different departments. If these movements are not properly documented, school personnel may have difficulty determining the current location or responsible user of an item. This can be particularly problematic for valuable

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equipment such as computers, projectors, printers, and other technology resources. Incomplete movement records can reduce accountability and make it difficult to identify missing or misplaced property [13].

Manual inventory processes can also contribute to delayed reordering. When stock levels are monitored through separate paper records or spreadsheets, personnel may not immediately recognize that an item is approaching its minimum required quantity. As a result, essential supplies may become unavailable before replacement items are purchased. Delayed procurement can disrupt classroom activities and administrative operations, particularly when the required materials are needed immediately. On the other hand, purchasing excessive quantities because of inaccurate or outdated inventory information may result in overstocking, unnecessary expenditures, and inefficient use of storage space [14].

Web-based inventory management systems provide an alternative approach by centralizing inventory information within a digital platform. Instead of maintaining separate paper records or independent spreadsheets, authorized users can record inventory information in a shared database. The system can maintain details such as item name, category, quantity, unit, condition, acquisition information, location, assigned personnel, and transaction history. Centralized records allow authorized personnel to access updated information through a web interface and reduce the need to search through multiple sources of information [14].

Digital inventory systems can also record different types of inventory transactions. For example, when a school receives new supplies, the quantity can be added to the available stock. When materials are issued to a classroom or employee, the transaction can be recorded and the available quantity adjusted accordingly. Returned items can also be documented to maintain an accurate record of their movement. By maintaining these transactions in a centralized system, schools can establish a more complete history of how resources are acquired, distributed, used, and returned [14].

Automated stock monitoring is another important feature of digital inventory management. A system can be configured with minimum stock levels for commonly used supplies. When the available quantity reaches or falls below the predefined threshold, the system can generate an alert or notification for authorized personnel. This allows school staff to identify items that may require replenishment before they are completely depleted. Such functionality can help reduce interruptions caused by shortages while also supporting more controlled purchasing decisions [14].

For educational institutions, asset management extends beyond monitoring the quantity of consumable supplies. Durable school assets require additional information because their value, condition, location, and maintenance requirements may change over time. A digital asset record

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can include information such as the date an item was acquired, its purchase reference, current condition, assigned department or user, physical location, maintenance history, and status. Maintaining these details allows administrators to monitor the complete lifecycle of school property from acquisition to eventual replacement or disposal [13].

Maintenance tracking is particularly important for equipment that requires periodic inspection or servicing. Computers, printers, projectors, laboratory equipment, and other school facilities may experience reduced performance or breakdown when maintenance is neglected. An inventory system can store maintenance schedules and records of previous repairs or servicing activities. Reminders can help responsible personnel identify equipment that is due for maintenance. Recording maintenance activities also creates a historical record that can assist administrators in determining recurring problems and evaluating whether continued repair or replacement is appropriate [15].

Asset assignment and location tracking can further improve accountability within the school. When equipment is assigned to a particular classroom, office, department, or authorized employee, the information can be recorded in the system. If the asset is transferred to another location, the transaction can be updated to reflect the change. This provides administrators with a clearer understanding of where school property is currently located and who is responsible for it. Such records can also make physical inventory checking more organized because personnel can compare the system's recorded location with the actual location of the asset [13].

Digital inventory systems can also support auditing activities. During an inventory audit, school personnel need to verify whether recorded items correspond to the physical resources available on school premises. A centralized database containing acquisition records, issuance transactions, returns, transfers, and current assignments can provide supporting information for this process. Historical records can help identify when an item was acquired, who received or used it, and whether it has been transferred or returned. This creates a more traceable inventory process and can improve accountability for school resources [15].

Inventory information can also contribute to school budgeting and procurement decisions. By examining historical consumption and transaction records, administrators can determine which supplies are frequently used and which assets require regular maintenance or replacement. This information can be used when preparing future procurement plans and allocating available funds. Instead of relying entirely on estimates, administrators can refer to documented inventory activity when determining the quantity and type of resources that may be required. This supports more evidence-based planning and can help reduce unnecessary purchases [15].

The integration of inventory management with procurement and financial functions can provide additional benefits. When purchasing information is connected with inventory records,

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administrators can compare items acquired with the quantities currently available and the rate at which resources are being consumed. This can help identify purchasing patterns and prevent unnecessary duplication of orders. Integration can also provide a clearer relationship between resource consumption and school expenditures, allowing administrators to monitor how available funds are being used for supplies and assets [14].

Another important consideration is the security of inventory information. School inventory records may contain information about valuable equipment, assigned personnel, storage locations, and purchasing activities. A web-based system should therefore provide appropriate access controls so that only authorized users can add, modify, or view specific inventory information. Role-based access can allow administrators to manage inventory records while giving other authorized personnel more limited functions, such as viewing available supplies or recording item transactions. Maintaining user accounts and transaction histories can also help identify changes made to inventory records and improve accountability.

For the proposed TCCIS School Management System, the principles identified in the literature can be applied through an integrated inventory and asset management module. The proposed system can allow authorized personnel to create and maintain records of school supplies, equipment, furniture, instructional materials, and other relevant assets. Each item can be associated with information such as its category, quantity, condition, location, assigned personnel, and transaction history. This centralized approach can reduce the school's dependence on separate paper records and spreadsheets while making inventory information easier to retrieve and update.

The proposed system can also support the recording of inventory transactions such as receiving, issuing, returning, and transferring items. When an item is issued, the system can update its available quantity and record the transaction. When an item is returned, the corresponding record can be updated again. These processes can provide a more consistent history of inventory movement and help school personnel determine the current status of resources. Automated reports can also assist administrators in reviewing available stock, frequently issued items, low-stock resources, and asset assignments.

By incorporating inventory management into the same platform as student records, attendance, grading, communication, and reporting, the proposed system can reduce the separation between different administrative functions. Authorized administrators can manage inventory information within the same centralized environment used for other school operations. This approach supports more organized record management and reduces the need to maintain multiple independent information sources. The integration of inventory data with the broader school management system can also provide a foundation for future improvements in procurement monitoring, asset maintenance, and resource planning [14].

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CHAPTER III

METHODOLOGY

3.1 Research Design

This study employs a developmental research design that combines qualitative and quantitative approaches throughout the planning, development, implementation, and evaluation phases of the proposed system [16]. Developmental research is appropriate for the study because its primary purpose is to design, develop, and evaluate a functional web-based school management system that addresses the identified administrative and academic needs of Timothy's Cradle Christian International School Inc. Rather than focusing solely on describing an existing situation, the research involves the systematic development of a technological solution based on actual requirements, observed processes, and feedback from intended users.

The qualitative component of the study is primarily used during the requirements-gathering and planning stages. Interviews, document analysis, and workflow observation are conducted to obtain detailed information about the school's existing procedures for managing student records, attendance, grades, inventory, communication, and reports. These methods allow the researchers to understand how school personnel currently perform their tasks, what information they require, which activities are repetitive or time-consuming, and what difficulties they encounter when using manual records, logbooks, and spreadsheets. The information gathered from these activities serves as a basis for identifying system requirements and determining the functions that should be incorporated into the proposed platform.

Interviews provide an opportunity for the researchers to obtain direct information from stakeholders regarding their experiences, needs, expectations, and concerns about the proposed system. Through guided questions, respondents can describe the current processes they follow and identify specific problems that may not be evident through document review alone. The information gathered from interviews can help the researchers determine which system features should be prioritized and how the proposed system should support actual school operations.

Document analysis is also included in the research design to examine existing forms, attendance records, student files, inventory records, reports, and other relevant documents used by the school. Reviewing these materials helps researchers understand the type of

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information currently collected, how records are organized, and what data fields or reporting requirements should be considered in the proposed system. Document analysis also provides a basis for designing database structures and system forms that correspond to existing school documentation while reducing unnecessary duplication.

Workflow observation is used to understand how administrative and academic tasks are performed in practice. By observing processes such as attendance recording, student information management, inventory monitoring, and report preparation, researchers can identify unnecessary steps, repeated encoding, delays, and other operational difficulties. Observation also helps ensure that the proposed system reflects actual work practices rather than assumptions about how school personnel perform their responsibilities. This is particularly important when developing a system intended to replace or improve manual procedures.

The development phase follows an iterative approach in which the identified requirements are translated into system features, interface designs, database structures, and functional modules. Instead of developing the entire system based on a single set of assumptions, prototypes can be created and reviewed during different stages of development. Feedback from intended users can then be used to identify unclear interfaces, missing functions, workflow problems, or other issues that require modification. This iterative process allows the system to gradually become more aligned with the operational requirements of the school [19].

Prototyping is particularly useful because it provides stakeholders with a visual and functional representation of the proposed system before the final version is completed. Users can interact with or review selected features and provide feedback regarding their usability, organization, and relevance to their responsibilities. Their comments can guide adjustments to navigation, forms, dashboards, data entry procedures, reports, and other system components. Through repeated cycles of development and feedback, the researchers can improve the system before final implementation and evaluation.

The quantitative component is applied during the evaluation phase to measure the performance and acceptability of the developed system. Evaluation may include structured assessment of characteristics such as functionality, usability, efficiency, reliability, security, accessibility, and overall user satisfaction. Responses gathered through evaluation instruments can be summarized using appropriate descriptive statistical measures to determine how users perceive the system and whether the developed features meet the identified requirements [16].

The combination of qualitative and quantitative methods allows the study to examine both the context in which the system will be used and the measurable performance of the developed solution. Qualitative methods provide detailed information about user experiences, existing workflows, and operational problems, while quantitative evaluation provides measurable evidence regarding the quality and acceptability of the developed system. Using both approaches strengthens the development process by ensuring that technical decisions are informed by actual user needs and that the completed system is evaluated using systematic criteria.

The research design also emphasizes continuous alignment between the identified requirements and the developed system. Information obtained during the initial investigation is translated into functional and non-functional requirements, which guide the design and development of the system. Feedback obtained during prototyping is then used to refine these requirements and improve system features. This process helps ensure that the final system is not based solely on theoretical assumptions but reflects the actual practices, limitations, and needs of the school [19].

Through this developmental research design, the study seeks to produce a practical and user-oriented web-based integrated school management system for Timothy's Cradle Christian International School Inc. The design provides a structured process for investigating existing problems, identifying requirements, developing the proposed solution, incorporating stakeholder feedback, and evaluating the resulting system. This approach supports the development of a system that is technically functional while also being relevant to the school's actual administrative and academic operations.

3.2 System Development Methodology

The Agile Software Development Model is adopted as the system development methodology for the proposed TCCIS School Management System. Agile is an iterative and incremental development approach that emphasizes continuous development, testing, evaluation, and improvement throughout the system development process [17]. This methodology is appropriate for the study because the proposed system contains several interconnected modules and requires continuous consideration of the needs and feedback of its intended users. Instead of developing the entire system at once and evaluating it only after completion, Agile allows the researchers to develop the system in smaller functional increments and make improvements throughout the development process.

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The Agile approach generally involves repeated cycles of planning, requirements analysis, system design, implementation, testing, review, and refinement. At the beginning of each iteration, the researchers identify the features and requirements that should be addressed. These requirements are then translated into interface designs, database structures, system functions, and other technical components. The selected features are implemented and tested before the results are reviewed. Issues identified during testing or feedback sessions can then be addressed in the next iteration. This cycle continues until the required system functions have been developed and refined [17].

The first stage of the development process focuses on analyzing the existing processes and identifying the requirements of the school. Information gathered through interviews, document analysis, and workflow observation is used to determine the problems associated with manual student records, attendance monitoring, inventory management, communication, and report preparation. The researchers then organize these findings into functional and non-functional requirements. Functional requirements describe what the system should be able to perform, while non-functional requirements address characteristics such as usability, security, reliability, accessibility, and performance.

After the requirements have been identified, the researchers proceed with system design. This includes designing the overall system architecture, user interfaces, database structure, navigation, user roles, and relationships among the different modules. The design process considers the responsibilities of different users, including administrators, teachers, and parents. Role-based access is incorporated into the design so that users can access only the information and functions relevant to their assigned responsibilities. The database design also establishes relationships between student records, attendance, grades, inventory transactions, communication records, and other relevant information.

The implementation phase develops the system features according to the approved requirements and design. Development is performed incrementally, beginning with the core components of the system. The student records module can be developed first because student information serves as a foundation for other functions. Once the basic student records functionality has been implemented and tested, subsequent iterations can introduce attendance monitoring, inventory management, parent–teacher communication, and reporting features. This incremental approach allows each major component to be developed and validated before additional functionality is introduced.

The student records module serves as the foundation for the succeeding modules because other system functions require reliable student information. The module can include student profiles, enrollment information, parent or guardian details, and other relevant records. During its iteration, the researchers can test functions such as adding, updating, viewing, searching, and managing student information. Feedback from users can be used to improve the

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organization of forms, search functions, navigation, and access permissions before the module becomes part of the larger integrated system.

The attendance module is then developed as another functional increment. It provides authorized teachers or staff with an interface for recording daily attendance and allows the system to store attendance information in the centralized database. The module can support attendance statuses such as present, absent, late, or excused, depending on the requirements identified during the analysis phase. Testing can determine whether attendance records are correctly associated with students, dates, classes, and authorized users. Any problems discovered during testing can be addressed through subsequent refinement.

The inventory module is developed in another iteration to support the management of school supplies, equipment, and other assets. The module can include functions for recording inventory items, quantities, item categories, issuance, returns, assignments, and other transactions. By connecting inventory information to the centralized database, authorized personnel can retrieve updated records and generate relevant inventory reports. User feedback can be incorporated to improve item categorization, transaction workflows, search functions, and reporting features.

The communication module is subsequently developed to support interaction between authorized school personnel and parents or guardians. This module can include announcements and direct messaging functions based on the requirements identified during the planning stage. The iterative approach allows the researchers to evaluate whether users can easily locate messages, compose communications, receive notifications, and access relevant announcements. Feedback from stakeholders can be used to improve the clarity and organization of communication features [19].

The reporting module is developed to consolidate information generated by the different system components. It can provide authorized users with reports related to student records, attendance, grades, inventory, and other relevant school information. Developing the reporting functions after the core modules have been established allows the system to generate reports using centralized and updated information rather than relying on manually consolidated records. The researchers can test whether reports contain accurate information, follow the required format, and can be generated efficiently.

Testing is performed throughout every development iteration rather than being postponed until the end of the project. Functional testing verifies whether individual features perform according to their requirements, while integration testing examines whether connected modules exchange and use information correctly. Usability testing can determine whether intended users can navigate the system and complete common tasks without unnecessary difficulty. Security-

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related testing can also examine whether role-based permissions prevent unauthorized users from accessing restricted functions or information.

Stakeholder feedback plays an important role in the Agile methodology. After a functional increment has been developed and tested, users can review the feature and provide comments regarding its functionality, usability, and alignment with actual school procedures. Feedback may identify missing features, unnecessary steps, confusing interfaces, or differences between the system workflow and existing school practices. The researchers can then prioritize these observations and incorporate appropriate changes into subsequent development iterations [19].

The iterative nature of Agile also allows changes to be accommodated during development. Requirements may become clearer as stakeholders interact with working prototypes and identify additional needs. Rather than treating the initial requirements as permanently fixed, Agile provides opportunities to revise selected features when changes are justified by user feedback or findings from testing. This flexibility is particularly useful for a school management system because actual workflows may contain details that are difficult to identify completely during the initial requirements-gathering stage.

Each iteration therefore follows a continuous cycle of planning, analysis, design, implementation, testing, review, and refinement. The completion of one iteration provides a functional increment that becomes the basis for the next stage of development. Through this process, the researchers can progressively build the integrated system while continuously checking whether each component meets the identified requirements. Problems can be identified and corrected earlier instead of accumulating until the final stage of development [17].

The Agile methodology also supports the integration of the different system modules. Although student records, attendance, inventory, communication, and reporting are developed as individual functional increments, they are designed to work together through a centralized database and consistent data structures. For example, student information entered into the student records module can be used by the attendance and reporting modules without requiring the same information to be repeatedly entered. This reduces duplication and supports consistency across the system.

For the proposed TCCIS School Management System, Agile provides a structured yet flexible methodology for developing the system according to actual stakeholder requirements. The iterative process enables the researchers to begin with essential functions and progressively introduce additional modules while continuously testing and refining each component. Through repeated stakeholder feedback and system evaluation, the methodology supports the development of a web-based platform that reflects the school's actual workflows and operational needs rather than relying solely on assumptions made during the initial planning stage [19].

3.3 System Architecture

The proposed TCCIS School Management System adopts a three-layer client–server architecture consisting of the Presentation Layer, Application Layer, and Data Layer [16]. This architecture separates the user interface, application processing, and data management responsibilities of the system. Such separation promotes better organization of the system components and makes it easier to maintain, test, update, and expand individual parts without unnecessarily affecting the other layers. The architecture also supports the integration of the system's major functions, including student records, attendance monitoring, academic records, inventory management, parent–teacher communication, and reporting.

Presentation Layer

The Presentation Layer represents the part of the system that users directly interact with. It is implemented as a responsive web interface using HTML5, Tailwind CSS, and JavaScript [24]. HTML5 provides the basic structure of the web pages, including forms, tables, navigation elements, dashboards, and other interface components. Tailwind CSS is used to organize the visual presentation and responsive layout of the system, while JavaScript supports interactive functions that improve the user experience.

The responsive design allows the system interface to adjust to different screen sizes and device types. Users can access the system through standard web browsers using desktop computers, laptops, tablets, or compatible mobile devices. This provides flexibility for teachers, administrators, and parents who may use different devices when accessing the system. The interface is designed to organize frequently used functions into accessible menus and dashboards, reducing unnecessary navigation and helping users locate relevant information more efficiently [24].

The Presentation Layer also provides different interfaces according to the user's assigned role. For example, administrators may have access to student management, inventory, reporting, and user management functions, while teachers may primarily access student information, attendance, grades, and communication features. Parents may be provided with access to authorized information concerning their children, such as attendance, academic information, announcements, and communication. This role-based presentation helps ensure that users are shown features appropriate to their responsibilities.

Forms within the Presentation Layer serve as the primary means of submitting information to the system. These may include student registration forms, attendance forms, grade-entry forms, inventory transaction forms, and communication forms. Input validation can be applied to help ensure that required information is entered in the correct format before it is processed by the

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Application Layer. Clear labels, organized fields, appropriate buttons, and consistent navigation are also considered to improve usability and reduce input errors.

Application Layer

The Application Layer contains the server-side logic responsible for processing requests received from the Presentation Layer. The proposed system uses PHP for server-side development. This layer acts as an intermediary between the web interface and the database, ensuring that user requests are processed according to the rules and requirements of the system.

The Application Layer is responsible for functions such as authentication, authorization, data validation, record creation, record updating, searching, report generation, and communication processing. When a user submits information through the web interface, the request is processed by the appropriate PHP logic before information is retrieved from or stored in the database. This separation prevents the Presentation Layer from directly managing database operations and provides a centralized location for implementing system rules.

Database interaction is performed using PHP Data Objects (PDO) and prepared statements [22]. Prepared statements separate SQL commands from user-supplied values, reducing the risk of SQL injection attacks. User inputs are handled through controlled database queries rather than being directly inserted into SQL statements. This provides an important security mechanism because the system manages sensitive information such as student profiles, attendance records, grades, parent information, and user account details.

The Application Layer also implements authentication and role-based access control. During login, the system verifies the user's credentials and determines the role associated with the account. Based on the assigned role, the application controls which functions and information the user is authorized to access. This prevents users from accessing administrative features or records that are outside their responsibilities. Authorization checks are applied to relevant requests rather than relying only on the visibility of interface elements.

Business rules are also implemented within the Application Layer. For example, attendance records must be associated with an appropriate student and date, grade records must be connected to the appropriate student and subject, and inventory transactions must update the corresponding stock information. These rules help maintain consistency across the different modules and ensure that information is processed according to the intended workflow.

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The Application Layer also facilitates communication between the different modules. Because the modules use a shared database structure, information entered into one part of the system can be retrieved by another authorized function. For instance, student information stored in the student records module can be used when recording attendance or grades. Similarly, attendance and academic information can be retrieved when generating reports. This reduces repetitive encoding and supports consistent information across the system.

Data Layer

The Data Layer is responsible for storing and managing the information used by the system. The proposed TCCIS School Management System uses MySQL as its relational database management system [23]. The database contains a normalized schema consisting of 18 interconnected tables organized into major functional groups, including core records, academic information, operational records, and public content.

The core records group contains fundamental information required by other system functions. This may include user accounts, student profiles, parent or guardian relationships, classes, and other foundational records. These tables provide the identifiers and relationships required to connect information throughout the system. For example, a student identifier can be used to associate a student's profile with attendance, grades, and other relevant records.

The academic group contains information related to academic activities, including subjects, attendance, and grades. Attendance records can be connected to individual students, teachers, classes, dates, and attendance statuses. Academic records can similarly be associated with students, subjects, school years, and the teacher responsible for recording the information. Establishing these relationships allows the system to retrieve and organize academic information without maintaining separate, disconnected records.

The operations group contains information related to administrative functions such as inventory and other school activities. Inventory-related tables can maintain information about items, quantities, assignments, and transactions. By storing inventory transactions in the database, the system can maintain a history of item movements and support the generation of inventory reports.

The public content group contains information intended for public-facing or general school communication functions, such as announcements and other website content. Separating public content from restricted administrative information helps distinguish information that may be displayed to general visitors from confidential records that require authenticated access.

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Database normalization is applied to reduce unnecessary duplication and improve data consistency [23]. Instead of storing the same information repeatedly in multiple tables, related information is separated into appropriate entities and connected through primary and foreign keys. For example, student information can be stored in a student table while attendance records reference the student through a corresponding identifier. This structure makes it easier to update information consistently and reduces the likelihood of conflicting duplicate records.

Relationships between tables also allow the different system modules to operate as an integrated platform. When a student record is created or updated, authorized modules can reference the same record instead of requiring users to enter the student's information repeatedly. This shared data structure supports the system's objective of reducing fragmented records and improving information consistency.

Client–Server Communication

The three layers communicate through a client–server process. When a user accesses the system through a web browser, the browser acts as the client and sends requests to the server. The server receives the request and processes it through the Application Layer. If information is required from the database, the application sends an appropriate query to the Data Layer. The resulting information is then processed by the application and returned to the Presentation Layer, where it is displayed to the user.

For example, when a teacher records attendance, the teacher enters the required information through the web interface. The request is sent to the PHP-based Application Layer, where the data is validated and processed. The application then uses a prepared database statement to store the attendance record in MySQL. After successful processing, the system can return a confirmation message to the teacher and make the updated attendance information available to other authorized functions.

This layered communication structure also allows changes to be made to one part of the system with reduced impact on the other layers. For example, improvements to the visual design of the web interface can generally be made within the Presentation Layer without changing the underlying database structure. Similarly, database optimization can be performed within the Data Layer while maintaining the same user interface and application functions. This separation supports maintainability and future system improvements [16].

Deployment Architecture

The proposed system supports two deployment configurations: cloud-based deployment through ProFreeHost and local-area-network deployment using XAMPP [22]. Cloud deployment

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allows authorized users to access the system remotely through an internet connection and a standard web browser. This configuration is appropriate for situations where administrators, teachers, or parents need to access authorized information from different locations.

Local-area-network deployment provides an alternative for environments where the system needs to operate within the school's local network. Using XAMPP, which provides a local Apache, PHP, and MySQL environment, the system can be hosted on a computer or local server and accessed by connected devices within the same network. This configuration can be useful for on-campus operations and development or testing activities where internet access is limited or unavailable.

The dual-deployment approach provides flexibility because the same general system architecture can support both remote and local access configurations. During development and testing, XAMPP can provide a controlled local environment for implementing and evaluating system functions. For remote accessibility, the system can be deployed to a web-hosting environment such as ProFreeHost, subject to the hosting service's capabilities and configuration requirements [22].

The deployment configuration also affects how users connect to the system. In cloud deployment, the client device communicates with the hosted web server through the internet. In local deployment, client devices communicate with the server through the school's local network. In both configurations, the Presentation, Application, and Data Layers remain logically separated, allowing the system's core functionality to remain consistent across deployment environments.

3.4 Tools and Technologies

The TCCISI School Management System uses HTML 5, Tailwind CSS, JavaScript, and AJAX for its responsive, interactive frontend. PHP 8.x powers the backend logic, while MySQL stores all system data securely. Developers use Visual Studio Code and Git for coding and version control, XAMPP for local testing with Apache, PHP, and MySQL, and Pro Free Host for cloud hosting that supports both PHP and MySQL.

LAYER	TECHNOLOGY
Frontend	HTML 5, tailwindCSS, JavaScript AJAX
Backend	PHP 8.x
Database	MySQL

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Development Environment	Visual Studio Code, Git
Local Hosting	XAMPP (Apache, PHP/MySQL)
Cloud Hosting	Pro Free Host (PHP/MySQL compatible)

3.5 Data Gathering Techniques

The study employs document analysis and semi-structured interviews as the primary data gathering techniques. These techniques are used to obtain relevant information about the existing administrative and academic processes of Timothy's Cradle Christian International School Inc. and to identify the requirements of the proposed TCCIS School Management System. The combination of documentary evidence and stakeholder perspectives allows the researchers to understand both the existing records and the actual experiences of the individuals involved in school operations [16][19].

Document Analysis

Document analysis involves the systematic review of existing school documents and records to determine the information currently collected, the format in which it is recorded, and the procedures followed in managing school activities [16]. The researchers examine relevant documents such as enrollment forms, student records, attendance logs, inventory ledgers, grade-related records, report templates, communication records, and other official school documents that are available and appropriate for the study.

The review of enrollment forms and student records helps the researchers identify the information required for student registration and profile management. These documents may contain fields such as student identification information, personal details, parent or guardian information, contact information, grade level, and other relevant enrollment data. Examining these fields provides a basis for determining which information should be included in the student records module of the proposed system.

Attendance logs are reviewed to determine how attendance information is currently recorded and organized. The researchers examine the information commonly included in attendance records, such as student name or identifier, date, class, teacher, and attendance status. The review also helps identify the procedures used to record absences, tardiness, or other attendance conditions. These findings can be used to design an attendance module that reflects the school's existing requirements while reducing repetitive manual recording.

Inventory ledgers and related asset records are examined to identify the information required for managing school supplies and property. The researchers may review fields such as item name, category, quantity, condition, acquisition information, assigned personnel, location, issuance, and return records. This information can guide the design of the inventory module and help determine the appropriate data fields and transaction processes required by the school.

Report templates and official school records are also analyzed to determine the information required for generating administrative and academic reports. The researchers examine the

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structure, required fields, organization, and frequency of reports to identify which system data should be included in automated reporting functions. This can reduce the possibility of developing reports that do not correspond to the school's actual documentation requirements.

Document analysis also helps the researchers identify similarities and differences among existing records. Some information may be recorded repeatedly across different forms or documents, which can indicate opportunities for centralization within the proposed system. Identifying duplicate information can help researchers design relationships between database tables and reduce unnecessary data entry. The review can also reveal inconsistencies in terminology, formats, or recording procedures that should be addressed during system design.

The information gathered through document analysis serves as a reference during database design, interface development, and workflow modeling. Rather than creating forms and system fields based solely on assumptions, the researchers can use existing school documentation as evidence of the information required in actual operations. This helps ensure that the proposed system reflects relevant school processes and documentation requirements [16].

Semi-structured Interviews

Semi-structured interviews are conducted to gather detailed information from individuals who are directly involved in or affected by the school's administrative and academic processes [19]. Unlike highly structured interviews that rely exclusively on predetermined questions, semi-structured interviews provide a consistent set of guiding questions while allowing researchers to ask follow-up questions when participants provide information that requires further clarification.

The participants may include school administrators, teachers, administrative staff, and parents, depending on their availability and relevance to the study. Each group can provide different perspectives regarding the existing procedures and the expected functions of the proposed system. Administrators can provide information about record management, reporting, monitoring, and administrative requirements. Teachers can describe their experiences with attendance recording, grade management, student information, and communication. Staff members can provide information about inventory and other operational activities, while parents can provide perspectives regarding access to student information, announcements, and communication with teachers.

During the interviews, participants are asked about their current procedures, difficulties encountered when performing their responsibilities, information they frequently need to access, and features they consider useful in a computerized system. Questions may also address problems related to manual record-keeping, delays in retrieving information, repetitive data

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entry, report preparation, inventory monitoring, and communication. Follow-up questions are used to clarify responses and obtain specific examples of existing workflows.

The interviews also help identify usability requirements for the proposed system. Participants can describe their expectations regarding navigation, accessibility, ease of use, forms, notifications, reports, and other interface features. Understanding these preferences is important because a technically functional system may still be difficult to adopt if users find its interface confusing or if its workflow does not correspond to their normal responsibilities [19].

Semi-structured interviews are also used to prioritize system features. Not every possible feature can necessarily be implemented within the scope of the project. Participants can therefore identify which functions are most important to their daily activities. The researchers can use these responses, together with findings from document analysis, to determine which features should receive greater attention during development.

Interview responses can also be compared with the findings from document analysis. For example, documents may indicate that certain information is recorded, while interviews may reveal difficulties in maintaining or retrieving that information. Comparing these sources provides a more complete understanding of the existing process and helps the researchers identify differences between documented procedures and actual practices.

The information collected from the interviews is documented, organized, and analyzed according to recurring themes and system requirements. Common concerns, requested features, workflow issues, and usability expectations are identified and considered during the requirements analysis and system design stages. Relevant findings are then translated into functional and non-functional requirements for the proposed TCCIS School Management System [19].

The use of both document analysis and semi-structured interviews provides complementary sources of information for the study. Document analysis establishes what information and formats are currently used by the school, while interviews provide insight into how these processes are experienced and performed by stakeholders. Combining these techniques helps the researchers develop a clearer understanding of existing workflows and provides a stronger basis for designing a system that responds to the actual administrative and academic requirements of Timothy's Cradle Christian International School Inc.

3.6 Testing Procedures

Testing is an essential part of the development of the proposed TCCIS School Management System because it determines whether the developed system functions according to the identified requirements and performs reliably under expected conditions. The testing process is conducted throughout development rather than only after the system has been completed. Different levels of testing are applied to identify functional errors, integration problems, usability concerns, compatibility issues, and security weaknesses. The study uses unit testing, system testing, and User Acceptance Testing (UAT) to evaluate the system at different stages of development [18][20].

Unit Testing

Unit testing involves testing individual functions, components, and modules independently before they are integrated into the complete system [18]. The purpose of unit testing is to determine whether each small component performs its intended function correctly based on its defined requirements. Testing individual components at an early stage makes it easier to identify and correct errors because problems can be isolated to a specific function rather than being investigated within the entire system.

For the student records module, unit testing may include checking functions such as adding a student, updating student information, searching for a student, viewing student details, and validating required fields. Each function is tested using appropriate inputs to determine whether the expected result is produced. Invalid or incomplete inputs can also be tested to determine whether the system responds appropriately and prevents incorrect information from being stored.

Attendance-related functions are also tested individually. These may include recording attendance, changing attendance status when authorized, retrieving attendance records, and generating attendance-related information. The researchers verify whether the correct student, date, class, and attendance status are associated with each record. Boundary and invalid-input cases can also be considered to determine whether the system handles unexpected entries appropriately.

For the inventory module, unit testing can be performed on functions such as adding inventory items, updating quantities, recording item issuance, recording returns, searching for items, and generating inventory information. The researchers verify that inventory quantities are correctly updated when transactions occur. For example, an item issued from available stock should result in the appropriate change to its recorded quantity.

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Other individual functions, such as user authentication, role assignment, grade recording, communication, notifications, and report generation, are also tested independently. Errors discovered during unit testing are corrected before the affected functions are integrated with other modules. This helps reduce the possibility of transferring unresolved errors into later stages of development [18].

System Testing

System testing evaluates the complete integrated system after the individual modules have been developed and combined. Unlike unit testing, which focuses on isolated functions, system testing examines whether the entire application operates correctly as a unified system [20]. The researchers perform end-to-end testing by following actual or representative workflows from the beginning of a process to its expected completion.

For example, a student record can be created through the student management module and subsequently used when recording attendance or grades. The researchers verify whether the information is correctly transferred between the relevant modules without requiring unnecessary duplicate encoding. Similarly, inventory information can be entered, updated through transactions, and subsequently reflected in inventory reports. These tests help determine whether the modules communicate correctly through the centralized database.

System testing also examines the major workflows performed by different user roles. Administrator workflows may include managing users, student records, inventory, reports, and system information. Teacher workflows may include viewing authorized student records, recording attendance, managing grades, and communicating with parents. Parent workflows may include viewing authorized student information, attendance, academic information, announcements, and communication features. Testing these workflows helps determine whether each user role can perform its intended activities successfully.

Cross-device and browser compatibility testing is conducted to determine whether the responsive web interface functions properly across different devices and screen sizes. The system can be tested using desktop computers, laptops, tablets, and compatible mobile devices. Different supported web browsers can also be used to identify display or functionality problems. Particular attention is given to navigation, forms, tables, buttons, dashboards, and other interface components that may behave differently depending on the device or browser.

System testing also includes security assessment. Authentication testing verifies whether only users with valid credentials can access protected areas of the system. Authorization testing examines whether users can access only the features and information permitted by their assigned roles. For example, a parent account should not be able to access administrative

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functions or another student's restricted information. These tests help determine whether the implemented role-based access controls operate as intended [20].

Data protection is also considered during system testing. The researchers verify whether sensitive information is transmitted and stored through appropriate mechanisms and whether unauthorized requests are rejected. Database interactions using prepared statements are also reviewed as part of the security assessment to reduce risks associated with unsafe database queries. The system is tested for common access-control and input-validation problems that could expose or improperly modify school information.

Error handling is another component of system testing. The researchers intentionally provide invalid, incomplete, or unexpected inputs to determine whether the system displays appropriate messages and prevents improper processing. Examples include submitting forms with required fields left blank, entering invalid values, attempting to access unauthorized pages, or performing actions with incomplete information. Proper error handling helps prevent incorrect records and provides users with understandable feedback when an operation cannot be completed.

User Acceptance Testing (UAT)

User Acceptance Testing is conducted to determine whether the developed system meets the practical needs and expectations of its intended users [20]. Unlike technical testing performed primarily by developers or researchers, UAT allows representative users to interact directly with the completed system and evaluate its functionality and usability based on realistic tasks.

Representative users may include administrators, teachers, and parents because these groups have different responsibilities and information requirements within the system. Administrators can evaluate administrative functions such as student management, inventory, user management, and reporting. Teachers can evaluate functions related to attendance, grades, student information, and communication. Parents can evaluate access to authorized student information, attendance, announcements, and communication features.

During UAT, participants are provided with representative tasks that reflect common activities they would perform when using the system. They may be asked to log in, navigate through the system, enter or view appropriate information, perform transactions, generate reports, or communicate through the available features. Their ability to complete these tasks provides information about the system's usability and suitability for actual school operations.

A structured feedback instrument based on the quality characteristics of ISO/IEC 25010 is used to collect user evaluations [20]. The evaluation may consider characteristics such as functional

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suitability, performance efficiency, compatibility, usability, reliability, security, maintainability, and portability, as applicable to the developed system. Users can provide ratings and comments based on their experience while performing the assigned tasks.

The feedback obtained during UAT is reviewed to identify features that require further improvement. Issues may include confusing navigation, unclear labels, missing functions, difficulties in completing tasks, slow responses, or other concerns identified by users. The researchers use these findings to determine whether modifications are necessary before the system is considered ready for its intended use.

UAT also provides an opportunity to determine whether the system reflects the workflows identified during the requirements-gathering stage. Users can compare the developed functions with their actual school procedures and identify differences that may not have been discovered during technical testing. This is particularly important for the proposed TCCIS School Management System because the system is intended to support existing administrative and academic activities rather than impose unnecessary changes on established workflows.

The results from unit testing, system testing, and UAT are documented throughout the testing process. Errors discovered during testing are recorded, analyzed, corrected, and retested to verify that the implemented solutions work as intended. Where user feedback identifies usability or workflow concerns, appropriate revisions are incorporated into the system. Repeated testing after corrections helps ensure that changes made to one function do not introduce new problems in other parts of the application [18][20].

The combination of these testing procedures provides multiple levels of evaluation for the proposed system. Unit testing verifies individual components, system testing evaluates the integrated application and its technical characteristics, and UAT determines whether representative users can effectively use the system for their intended tasks. Together, these procedures provide a structured basis for identifying defects, improving usability, validating security controls, and determining whether the TCCIS School Management System satisfies its defined functional and quality requirements.

3.7 Evaluation Criteria

The evaluation criteria are established to determine whether the developed system meets the expected technical and operational requirements in terms of performance, reliability, usability, functionality, and accessibility. These criteria provide measurable indicators for assessing how effectively the system performs its intended functions under normal operating conditions and during simultaneous use by multiple users. The evaluation results will serve as a basis for identifying system strengths, detecting areas that require improvement, and determining

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whether the system is suitable for actual use within Timothy's Cradle Christian International School Inc. [20].

METRIC	BENCHMARK
Response Time	2 Seconds for priority operations
Concurrent Users	50 simultaneous users with stable performance
System Uptime	~99.9% during operational hours

Performance evaluation will focus on the responsiveness and efficiency of the system when users perform priority operations such as logging in, searching student records, recording attendance, entering grades, updating inventory, sending messages, and generating reports. The target response time for priority operations is set at two seconds or less, ensuring that users can complete routine tasks without unnecessary delays. Response-time testing will be conducted under normal operating conditions and will consider the time required for the system to process user requests and display the corresponding results. Operations that consistently exceed the established threshold will be reviewed to identify possible causes such as inefficient queries, excessive data processing, or network-related delays.

The system will also be evaluated for its ability to maintain stable performance when accessed by multiple users simultaneously. A minimum benchmark of 50 concurrent users will be used to assess whether the system can continue processing requests without significant performance degradation, unexpected errors, or service interruption. This criterion is particularly important because administrators, teachers, and parents may access different system functions at the same time. Load and stress testing will therefore be used to observe the system's behavior under increasing numbers of simultaneous requests and to identify potential limitations in server resources, database processing, or application performance.

System availability and reliability will also form part of the evaluation. The target is approximately 99.9% system availability during operational hours, indicating that the system should remain accessible and functional when users need to perform school-related activities. Availability testing will consider occurrences of system downtime, failed requests, unexpected errors, and interruptions that may prevent users from accessing important functions. Any identified service interruptions will be documented and analyzed to determine their causes and possible corrective measures [20].

In addition to technical performance, the system will be evaluated based on functional suitability, usability, reliability, security, compatibility, and accessibility. Functional suitability will determine whether the implemented modules perform the required tasks correctly and completely. Usability will assess whether administrators, teachers, and parents can understand and

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navigate the system with minimal difficulty. Reliability will examine the consistency of system operations and its ability to maintain correct performance over repeated use. Security evaluation will focus on authentication, role-based authorization, protection of user information, and secure handling of database transactions. Compatibility and accessibility will assess whether the system functions properly across supported browsers, desktop computers, and mobile devices.

The evaluation results will be recorded and compared against the established criteria. Findings from performance testing, system testing, and User Acceptance Testing will be used to determine whether further improvements are necessary. Issues identified during evaluation will be documented, corrected, and retested to ensure that the final version of the system satisfies the defined technical and user requirements before deployment.

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CHAPTER IV

RESULTS AND DISCUSSION

4.1 System Overview

The completed Timothy's Cradle Christian International School Management System (TCCIS SMS) is a web-based integrated school management system designed to support and streamline the administrative, academic, and communication activities of Timothy's Cradle Christian International School Inc. The system brings together three interconnected components: a role-based internal portal, a public information website, and a centralized database. These components operate using a unified codebase and consistent data model, allowing information to be organized, maintained, and accessed according to the authorized functions of each user [24]. The integrated structure reduces the need for separate and disconnected systems while providing a centralized environment for managing school information.

The public information website serves as the school's online information platform for students, parents, prospective families, and other visitors. It provides important school-related information without requiring users to access the internal management portal. The website includes school announcements, academic and school calendars, enrollment information, downloadable forms, staff directory, frequently asked questions (FAQs), and other general school information [22]. By presenting these resources in one accessible location, the public website helps users obtain commonly requested information without relying entirely on direct inquiries or printed materials. The website can also support the school's communication and information-dissemination activities by providing updated announcements and relevant notices.

The role-based internal portal serves as the primary workspace for authorized school users. Access to the portal is controlled according to the user's assigned role, ensuring that users are provided with functions and information appropriate to their responsibilities. Administrators can manage student information, user accounts, attendance records, academic records, inventory, communications, and reports. Teachers can access functions related to their assigned students and classes, including attendance recording, grade management, student information, and parent communication. Parents can access relevant information about their children, including attendance, academic progress, announcements, and communication features. Role-based access helps prevent unauthorized users from viewing or modifying information outside their assigned responsibilities.

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The system's centralized database serves as the primary storage layer for information generated and managed by the different system components. Student profiles, user accounts, class information, attendance records, grades, inventory records, announcements, messages, and other relevant information are stored in a structured relational database. The centralized approach allows authorized modules to retrieve updated information from a common data source, reducing unnecessary duplication and inconsistencies between different parts of the system. Relationships between database tables also allow information to be connected across modules, such as linking students with their classes, attendance records, academic records, and parent accounts.

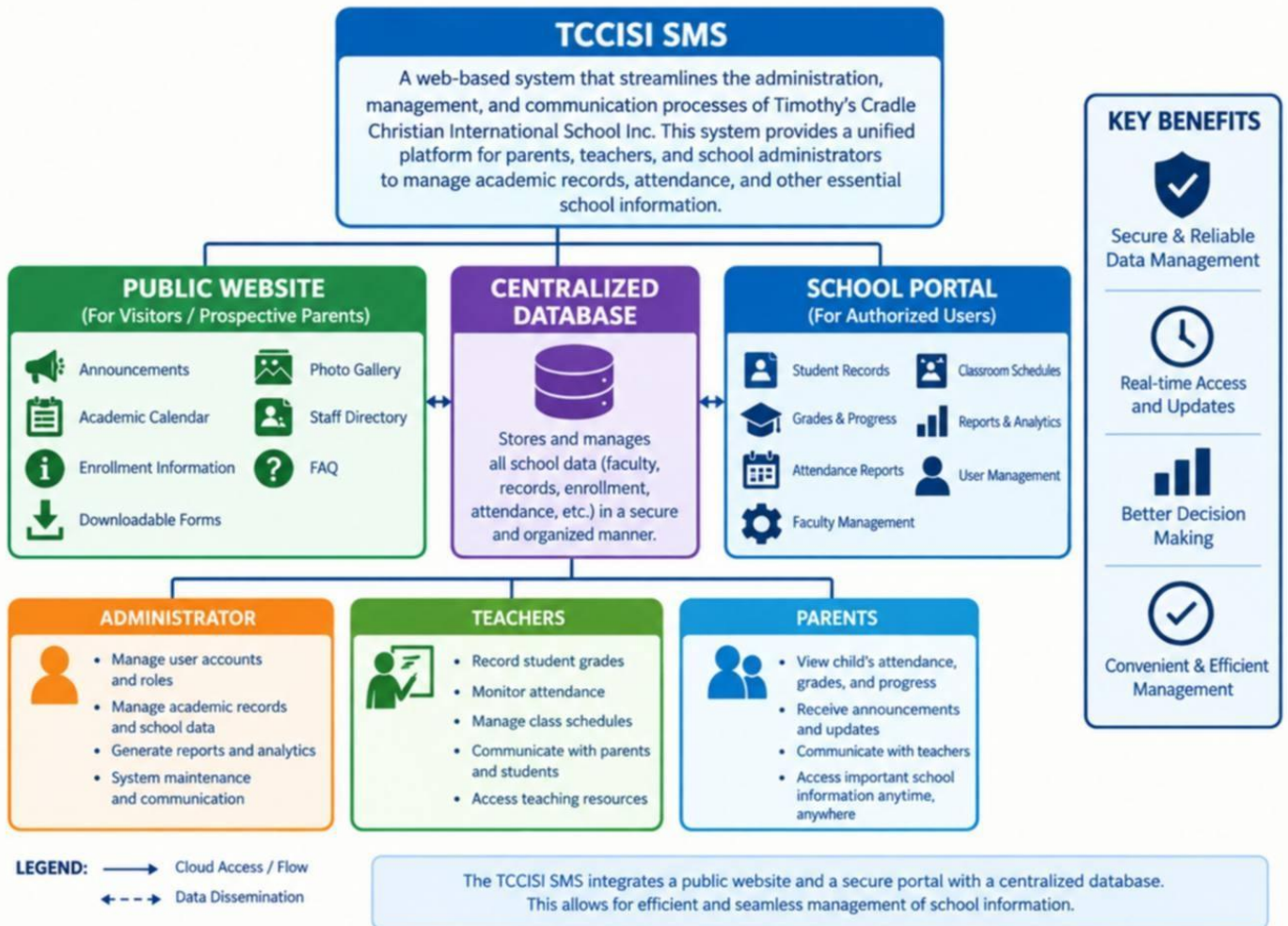
The three components are designed to work together while maintaining appropriate access boundaries. Information managed through the internal portal is connected to the centralized database, while selected school information can be presented through the public website. For example, administrators may update school announcements or calendar information through the management environment, after which the appropriate information can be displayed on the public-facing website. This approach reduces repetitive data entry and helps ensure that published information remains consistent with the information maintained by authorized school personnel.

The system also supports the digitization of several processes that were previously handled through physical documents, logbooks, and spreadsheets. Instead of maintaining separate records for different activities, authorized users can perform tasks through the corresponding modules within the system. Attendance can be recorded electronically, student information can be searched and updated through centralized records, inventory transactions can be monitored digitally, and reports can be generated from stored system data. These functions help reduce repetitive administrative work and improve the accessibility and organization of school information.

Overall system operation follows a controlled flow in which users access the appropriate interface, perform authorized activities, and interact with the centralized database through the application layer. Authentication and role-based authorization are applied to protect restricted functions and information. The architecture also allows the different modules to communicate with shared data while maintaining separation between public information and internal school records. Through this integrated design, TCCIS SMS provides a unified platform that supports school administration, teaching activities, parent communication, information dissemination, and record management within a single web-based environment [22][24].

OVERVIEW OF THE DEVELOPED SYSTEM

Timothy's Cradle Christian International School Inc (TCCISI SMS)



4.2 Key Functional Modules

The completed TCCIS SMS consists of several interconnected functional modules designed to support the major administrative, academic, operational, and communication requirements of Timothy's Cradle Christian International School Inc. Each module performs a specific set of functions while sharing information through the centralized database. The integration of these modules allows authorized users to perform school-related tasks within a single system rather than relying on separate paper records, spreadsheets, and external communication platforms.

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Student Records Management. The Student Records Management module provides a centralized digital environment for storing and maintaining student information. It supports digital enrollment forms, student profile creation, profile updating, record searching, and academic record archiving [2]. During enrollment, authorized personnel can enter required student information into standardized digital forms, reducing the need for repeated manual encoding. The system can automatically create a student profile after the required information has been recorded, allowing the student's information to become immediately available to authorized users.

The module maintains important information such as student identification details, personal information, parent or guardian information, class assignment, and relevant academic records. Authorized users can search and retrieve records using available identifiers or filtering options, reducing the time required to locate information compared with searching through physical filing cabinets. Updates can also be recorded directly in the system, helping maintain more organized and consistent student information. Academic records can be retained digitally, providing a structured record that can be referenced when preparing reports or reviewing a student's school history [2].

Attendance Tracking. The Attendance Tracking module provides an electronic method for recording and monitoring daily student attendance. Teachers or authorized personnel can select the appropriate class and mark each student's attendance status, such as Present, Absent, Late, or Excused. Attendance records are associated with the corresponding student, class, date, and recording user, allowing the system to maintain an organized history of attendance activities [9].

The system automatically calculates attendance percentages based on recorded attendance data, reducing the need for manual computation. It can also support configurable absence or tardiness alerts so that designated users can identify attendance concerns more quickly. When connected to the communication portal, relevant attendance notifications may be communicated to parents on the same day. This reduces the time required to compile attendance records manually and provides teachers and administrators with more accessible attendance information for monitoring student participation and identifying recurring attendance concerns [9].

Grade Management. The Grade Management module is designed to assist teachers and authorized personnel in recording and maintaining student academic performance. The module supports quarterly or term-based grade encoding, progress computation, term closing, and report generation [6]. Teachers can enter grades according to the assigned subjects, students,

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grading period, and academic year. The system associates each grade with the appropriate student and subject, helping maintain organized academic records.

Built-in validation mechanisms are used to reduce common data-entry errors, such as incomplete entries, invalid values, or incorrectly assigned records. The module can also perform automated calculations required for progress monitoring and reporting, reducing repetitive manual computation. Once grades have been reviewed and finalized for a grading period, the information can be used in generating academic reports and other required documents. Centralized grade storage also helps maintain consistency when academic information is accessed by authorized users across different sections or system functions [6].

Inventory Control. The Inventory Control module provides a digital mechanism for managing school supplies, equipment, materials, and other assets. Authorized personnel can register inventory items and record relevant information such as item name, category, quantity, condition, location, and other applicable details. The module also records inventory transactions, including the issuance, return, addition, or adjustment of items [14].

The system provides a current stock display based on recorded transactions, allowing authorized personnel to monitor available quantities more efficiently. Low-stock alerts can be configured to identify items that have reached a predefined minimum quantity. This feature helps personnel recognize potential shortages before they significantly affect school operations. Transaction histories also provide an accountability record by showing changes in inventory and supporting the monitoring of item movement. By maintaining centralized and updated inventory information, the module can support more timely procurement and resource-management decisions [14].

Communication Portal. The Communication Portal provides a centralized environment for communication between authorized school users and relevant stakeholders. It includes role-based announcement boards for publishing general school information and private direct messaging for authorized one-to-one communication [8]. Administrators can publish announcements, notices, reminders, and other school-related information, while teachers and parents can communicate through the appropriate messaging functions.

Access to communication features is controlled according to user roles to ensure that users can only access functions and conversations permitted by the system. Message and announcement records can be maintained as part of an audit trail, providing a history of communication activities within the platform. This reduces dependence on external messaging applications or SMS services for routine school communication and provides a more organized location for maintaining school-related conversations. The centralized communication environment also

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helps users locate previous messages and announcements when information needs to be reviewed.

Reporting Engine. The Reporting Engine provides automated tools for generating school reports based on information stored in the centralized database [5]. Authorized users can select relevant criteria and filters before generating reports, allowing outputs to focus on specific students, classes, dates, grading periods, inventory categories, or other applicable parameters. The module supports the generation of documents such as attendance summaries, grade sheets, master lists, and inventory status reports.

Generated reports can be exported into PDF and CSV formats, allowing information to be viewed, printed, stored, or further processed when necessary. Automated report generation reduces repetitive manual compilation and computation because the system retrieves information directly from existing records. This also helps maintain consistency between stored data and generated reports. By providing on-demand access to organized reports, the Reporting Engine supports administrative monitoring, academic documentation, inventory management, and decision-making activities within the school [5].

The integration of these functional modules allows TCCIS SMS to operate as a unified school management platform. Information entered in one module can support related functions in other modules through the centralized database. For example, student records provide the information needed for attendance and grade management, attendance data can be included in reports and communication activities, and inventory transactions can be transformed into inventory status reports. This interconnected design minimizes duplicate data entry and allows authorized users to manage related school processes within one system.

4.3 Performance Outcomes

The performance evaluation of the completed TCCIS SMS was conducted to determine whether the developed system could perform its intended functions efficiently, reliably, and securely under the defined operating conditions. The evaluation focused on system response time, data retrieval, concurrent access, reliability, security, usability, and alignment with the actual workflows of Timothy's Cradle Christian International School Inc. The results provide an indication of how effectively the implemented system addresses the operational difficulties associated with manual and fragmented record-management processes.

The system demonstrated satisfactory performance in terms of response time. During standard operating conditions, system response times averaged approximately 1.2 seconds, which is within the established 2-second performance benchmark. This result indicates that common operations such as logging in, opening student records, searching information, recording attendance, updating inventory records, and accessing reports could be completed without significant waiting time. Faster response times are particularly important for routine school activities because teachers and administrative personnel may perform multiple transactions throughout the day. Maintaining responsive system behavior helps reduce interruptions and allows users to complete their assigned tasks more efficiently.

A significant improvement was also observed in data retrieval time. Under the previous manual process, locating specific student or administrative information could require approximately 5–15 minutes, depending on the location and organization of physical records. With the TCCIS SMS, relevant information can be searched and retrieved from the centralized database in under 3 seconds [2]. The digital search capability reduces the need to manually inspect filing cabinets, logbooks, or separate spreadsheet files. It also allows authorized users to retrieve information using available search fields and filters, making frequently accessed records easier to locate.

The improvement in retrieval time also contributes to more efficient record management and administrative workflow. Information stored in the centralized database can be accessed by authorized users without requiring physical movement of documents between offices or personnel. This reduces delays when records are needed for attendance verification, grade preparation, report generation, enrollment-related activities, or other administrative tasks. Centralized storage also provides a more structured approach to maintaining records because related information can be connected through the system's database relationships.

The system was additionally evaluated based on its ability to support multiple users and concurrent operations. The performance test used the established benchmark of supporting up to 50 concurrent users while maintaining stable system operation. The evaluation considered

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whether simultaneous requests could be processed without significant system slowdown, failed transactions, or unexpected interruption. Stable operation under concurrent access is important because administrators, teachers, and parents may access different parts of the system at overlapping times. The result of this testing provides evidence of the system's ability to accommodate expected usage within the institution's operational environment.

Security and access control were also considered important performance outcomes. The implemented role-based access control successfully restricted sensitive information and system functions according to the user's assigned role [20]. Administrators, teachers, and parents were provided with access appropriate to their responsibilities, preventing unauthorized users from accessing restricted records or management functions. This approach is particularly important for student information because the system stores academic, attendance, personal, and communication-related data.

The system also maintained activity and modification records for accountability. Relevant changes made within the system were logged, allowing authorized personnel to identify and review system activities when necessary [20]. Maintaining an audit trail provides a record of important modifications and supports monitoring of data changes. It can also assist administrators in investigating discrepancies and determining when and through which authorized account a particular modification was made.

The User Acceptance Testing (UAT) results further indicated that users responded positively to the system's usability and practical functionality [19]. Feedback from representative users indicated that the system was understandable, easy to navigate, and aligned with common daily school workflows. The organization of the modules allowed users to locate relevant functions without relying on separate applications for each activity. The feedback also provided opportunities to identify interface elements and processes that could be refined during the development cycle.

The system's usability outcome is particularly relevant because the project was developed specifically around the operational requirements of the institution. Rather than adopting a generic platform with unnecessary functions, the TCCIS SMS focuses on processes directly related to student records, attendance, grades, inventory, communication, and reporting. This custom-built approach allowed system functions and workflows to be designed around the identified needs and constraints of the school [19].

The evaluation also demonstrated improvements in workflow integration. Student information stored in the Student Records Management module can support attendance and grade management, while attendance and academic information can be used in generating reports. Inventory transactions can similarly be converted into updated inventory status reports, while

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the communication module provides a centralized channel for school-related announcements and messages. The integration of these functions reduces repeated encoding and allows related information to be accessed through interconnected modules.

The results therefore demonstrate measurable improvements in several areas of system operation. The 1.2-second average response time satisfied the established response-time requirement, while under-three-second data retrieval represented a substantial reduction from the previous 5–15-minute manual search process. Role-based access and activity logging provided mechanisms for controlling and monitoring access to sensitive information [20]. User acceptance feedback also indicated that the system's usability, clarity, and workflow alignment were appropriate for the intended users [19]. These outcomes provide the basis for evaluating the completed TCCIS SMS against the technical and operational requirements established during the development process.

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CHAPTER V

CONCLUSION AND RECOMMENDATIONS

Introduction

This chapter presents the summary of findings, conclusions, and recommendations of the study entitled “A Web-Based Integrated School Management System for Timothy’s Cradle Christian International School Inc.” The study was conducted to design and develop a centralized web-based system that would address the challenges encountered by the school in managing its academic, administrative, and operational activities. The development of the system was based on the identified needs and existing processes of the school, particularly those involving student records, attendance monitoring, grade management, inventory management, communication, and report generation.

The study recognized that traditional and manual methods of managing school information may require considerable time and effort from school personnel. The use of paper-based records, spreadsheets, logbooks, and other separate methods of recording information can make data retrieval and updating more difficult. These processes may also increase the possibility of errors, duplication of information, misplaced records, and delays in preparing reports. In addition, the absence of a centralized platform may make communication between teachers, parents, and school administrators less organized and less efficient.

To address these concerns, the researchers developed a web-based integrated school management system that brings several essential school functions into a single platform. The system provides authorized users with appropriate access to student information, attendance records, grades, inventory data, communication features, and generated reports. Through role-based access, users can perform functions according to their responsibilities while helping maintain the privacy and security of school information. The integration of these functions is intended to improve the organization of data and make important information easier to access and manage.

The development of the system followed a systematic process that included requirements identification, planning, system design, development, testing, implementation, and evaluation. During the development process, the researchers considered the requirements and concerns identified from the existing school processes. These requirements served as the basis for designing the system’s features, database structure, user interface, and access controls. The developed system was then subjected to testing to identify functional errors, usability concerns, and other issues that could affect its performance.

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The findings presented in this chapter are based on the results obtained during system development, functional testing, performance evaluation, and User Acceptance Testing (UAT). These evaluation activities were conducted to determine whether the developed system was capable of performing its intended functions and whether it was acceptable to its intended users. The evaluation also provided information regarding the system's ability to support school personnel in completing common tasks more efficiently and systematically.

The summary of findings focuses on the extent to which the developed system achieved the objectives established in the study. It discusses the results related to the management of student records, attendance, grades, inventory, communication, and report generation. It also considers the system's performance in terms of important software quality characteristics such as functionality, usability, reliability, efficiency, security, and overall user acceptability. The findings provide an objective basis for determining the effectiveness of the developed system in addressing the identified problems.

The conclusions presented in this chapter are drawn directly from the findings and evaluation results. They explain how the developed system responded to the problems identified at the beginning of the study and whether the objectives of the research were achieved. The conclusions also emphasize the potential contribution of the system to improving the school's information management processes and supporting users in their daily responsibilities.

Finally, the recommendations provide possible improvements that may be considered for future versions of the system. These recommendations are based on the limitations identified during the development and evaluation of the system, as well as possible future needs of the school. Potential enhancements may include improving system accessibility, strengthening security and data backup mechanisms, adding additional academic and administrative features, improving communication capabilities, and providing support for future integrations.

Overall, this chapter provides a comprehensive summary of the results of the study and establishes the basis for determining the effectiveness and applicability of the Web-Based Integrated School Management System for Timothy's Cradle Christian International School Inc. The findings, conclusions, and recommendations presented herein serve as the final assessment of the developed system and provide direction for its possible improvement and future implementation.

5.1 Summary of Findings

The study resulted in the successful design and development of the Timothy's Cradle Christian International School Management System (TCCIS SMS), a web-based school management system developed to address the identified challenges in the school's existing administrative and academic processes. The system was designed to provide a centralized platform where authorized users could manage student information, attendance, grades, inventory, communication, and reports. The development of the system focused on reducing the school's dependence on manual and paper-based procedures and providing a more organized, accessible, and efficient way of managing school information.

The first finding of the study was the successful implementation of the Student Attendance Management module. The system enabled teachers and authorized personnel to record student attendance electronically rather than relying solely on handwritten attendance sheets. Attendance records could be stored in the database and retrieved whenever needed. The system also maintained attendance histories, allowing authorized users to review a student's attendance information over a specific period. This feature helped make attendance monitoring more organized and reduced the possibility of losing or misplacing attendance records. It also minimized repetitive work involved in manually compiling attendance information for reports and monitoring purposes.

The Student Records Management module was also successfully developed and integrated into the system. This module provided a centralized database for storing and managing essential student information. Authorized users could add new student records, update existing information, search for specific students, and retrieve records when needed. The centralized storage of information reduced the need to search through multiple physical files or separate documents. Student information could also be connected with related attendance, academic, and other school records, allowing authorized users to access relevant information from a single system. This helped improve the organization and accessibility of student data.

The study also found that the Inventory Control module provided a structured approach to managing school supplies, equipment, materials, and other resources. Authorized users could record inventory items and maintain information regarding their quantities and availability. The system also allowed users to monitor inventory transactions, including the movement or changes in available items. By maintaining updated inventory information, the school could more easily identify items that were becoming limited and determine which resources needed replenishment. The module therefore provided a more systematic alternative to manually checking and recording inventory information.

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The Communication Portal was another important component successfully incorporated into the TCCIS SMS. The module provided authorized users with a centralized platform for sharing school-related announcements and messages. Instead of depending entirely on different communication platforms, users could access relevant information through the school management system. Announcements and messages could be organized within the platform, making it easier for users to locate and review school-related information. The centralized communication feature also supported the timely distribution of important notices and helped provide a more organized method of communication between authorized users.

The Reporting Engine successfully supported the generation of different school reports required for administrative and academic purposes. Authorized users could generate reports such as attendance summaries, grade sheets, master lists, and inventory reports based on information stored in the system. The reporting functionality reduced the need to manually collect information from different sources and repeatedly prepare reports using separate documents. Since the reports were generated from centralized system records, users could obtain organized information more efficiently. This feature also helped reduce the amount of time spent on repetitive report preparation and allowed school personnel to focus on other administrative responsibilities.

The system also demonstrated satisfactory technical performance during testing and evaluation. The recorded average response time was approximately 1.2 seconds, which was within the established two-second benchmark. This result indicated that the system was able to process common user requests within an acceptable period. Frequently needed information could also be retrieved in less than three seconds. This represented a considerable improvement compared with the previous manual retrieval process, which required approximately five to fifteen minutes to locate and prepare certain information. Faster retrieval allowed authorized users to access necessary records more conveniently during school operations.

The results of the User Acceptance Testing (UAT) further indicated that the developed system was understandable and usable for its intended users. During the testing process, users were able to interact with the different modules and perform tasks related to their assigned responsibilities. The system's interface and functions were evaluated based on their ability to support common school activities. The results indicated that the major functions of the system were understandable to the users and that the system generally corresponded with the existing workflows and information management requirements of the school.

The implementation of role-based access control provided an additional mechanism for managing access to school information. Users were assigned appropriate roles, which determined the functions and information they were permitted to access. This helped prevent unauthorized users from accessing functions that were not relevant to their responsibilities. The system also incorporated activity logging, which provided a record of selected user activities

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performed within the system. This feature supported monitoring and accountability by allowing authorized personnel to review system activities when necessary.

The integration of these modules provided the school with a centralized environment for performing different school management activities. Instead of managing attendance, student records, inventory, communication, and reports through separate manual procedures, the developed system brought these functions together within one web-based platform. The use of a shared database also helped maintain consistency among related records because information entered into the system could be used by appropriate modules for different school processes.

The findings also showed that the system reduced several repetitive administrative tasks. Electronic attendance recording reduced the need to manually compile attendance information, centralized student records reduced the time required to locate files, inventory monitoring provided updated information about available resources, and automated reporting reduced repetitive report preparation. These functions contributed to a more organized workflow for authorized users and provided easier access to information needed for daily school operations.

The development and testing of the TCCIS SMS demonstrated that the proposed system was capable of supporting the identified school management requirements. The implemented modules addressed key areas of school operations, while the performance testing and User Acceptance Testing provided evidence regarding the system's responsiveness, usability, accessibility of information, and alignment with the intended workflows of the school.

5.2 Conclusion

The development of the Timothy's Cradle Christian International School Management System (TCCIS SMS) successfully addressed the identified challenges associated with the school's reliance on manual, paper-based, and fragmented processes. The study demonstrated that integrating major administrative, academic, operational, and communication activities into a single web-based platform can improve the organization, accessibility, and management of school information. Through the developed system, processes that previously depended on physical records, logbooks, spreadsheets, and separate communication methods were transformed into centralized digital workflows.

The system successfully achieved the stated objective of automating student attendance management. Teachers and authorized personnel can record daily attendance electronically, maintain historical attendance records, and access attendance information more efficiently. Automated calculations reduce the need for repetitive manual computation, while organized attendance records allow authorized users to monitor student attendance more conveniently.

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The availability of structured attendance information also supports the preparation of attendance reports and provides a more reliable basis for monitoring attendance-related concerns.

The objective of centralizing student records was also achieved through the Student Records Management module. Student information that was previously maintained through physical documents and separate files can now be stored within a centralized database. The system provides authorized users with search and retrieval functions, reducing the time required to locate important information. The centralized structure also establishes relationships between student profiles, classes, attendance, academic records, parent accounts, and other relevant information, creating a more organized environment for managing student data.

The system also achieved the objective of developing an inventory tracking mechanism. Through the Inventory Control module, authorized personnel can register items, monitor available quantities, record inventory transactions, and identify items requiring replenishment. Digital transaction records provide greater visibility of item movement and support accountability for school resources. The availability of low-stock information can also assist personnel in identifying potential shortages and making more timely procurement decisions.

Another major objective achieved was the establishment of a centralized communication platform for school stakeholders. The Communication Portal provides announcement and direct-messaging functions that allow authorized users to communicate through the system. This reduces the need to depend entirely on separate communication applications for school-related information and provides a more organized location for maintaining relevant announcements and conversations. Role-based access also ensures that communication functions are provided according to the responsibilities and permissions of each user.

The objective of automated report generation was likewise achieved. The Reporting Engine allows authorized users to generate organized reports based on information stored in the centralized database. Attendance summaries, grade sheets, master lists, and inventory reports can be generated according to applicable filters and exported into supported formats. This reduces repetitive manual compilation and helps ensure that reports are based on current information stored within the system. As a result, school personnel can access required information more efficiently for internal management, documentation, and reporting requirements [26].

The system's performance evaluation further demonstrated that the developed solution met the established technical expectations. The recorded average response time of approximately 1.2 seconds was within the defined two-second benchmark, while frequently required information could be retrieved in under three seconds compared with the previous manual retrieval time of approximately five to fifteen minutes. These results demonstrate an improvement in the speed

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and accessibility of school information. The implementation of role-based access and activity logging also provided mechanisms for protecting sensitive information and maintaining accountability for system activities.

The findings from User Acceptance Testing further indicated that the system was understandable, usable, and aligned with the daily workflows of the intended users. The modular design allowed administrators, teachers, and parents to access functions appropriate to their respective roles. By focusing on the specific processes and requirements identified within the institution, the system avoided unnecessary functions that could increase complexity and instead provided features directly related to the school's identified operational needs [27].

The study also demonstrates the value of a purpose-built and affordable web-based solution for educational institutions with limited resources. Rather than requiring a large-scale commercial platform with features that may exceed the needs of a small institution, TCCIS SMS provides a focused set of functions designed around the school's actual administrative and academic processes. The centralized architecture, role-based access, integrated modules, and automated reporting capabilities provide a foundation for continued digitalization while allowing the institution to manage its resources more efficiently [27].

Despite these achievements, the system remains dependent on appropriate technical infrastructure, accurate user input, reliable connectivity for web-based access, and proper system administration. These factors should be considered when implementing and maintaining the system in an actual operational environment. Continued maintenance, user training, data backup, security monitoring, and future system improvements will be important to sustain the benefits achieved through the project.

In conclusion, the TCCIS SMS demonstrates that the identified administrative and academic processes can be effectively transformed from fragmented manual procedures into an integrated digital environment. By centralizing records, automating routine activities, improving information retrieval, supporting communication, managing inventory, and generating reports, the system provides a practical technological solution to the operational requirements identified in the study. The successful achievement of the project's objectives supports the use of a customized web-based school management system as a foundation for improving the efficiency, organization, accessibility, and accountability of school operations at Timothy's Cradle Christian International School Inc. [26][27]

5.3 Recommendations

Based on the findings, evaluation results, and identified limitations of the completed TCCIS SMS, several recommendations are proposed to further improve the system's functionality, reliability, scalability, and usefulness to Timothy's Cradle Christian International School Inc. These recommendations are intended to guide future system enhancements and provide possible directions for subsequent development.

Enhanced Attendance Verification. Future versions of the system may incorporate additional attendance verification mechanisms, such as QR-code scanning or biometric authentication, to complement or replace manual attendance selection [9]. A QR-code-based approach could allow teachers or authorized personnel to record attendance by scanning a student-specific or session-specific code, reducing the number of manual selections required during attendance-taking. Biometric authentication could provide another method of identity verification by using unique biological characteristics. If implemented, these technologies should be evaluated carefully in terms of cost, privacy, security, usability, and suitability for the school's environment. Appropriate safeguards should also be established to protect attendance-related personal information.

Offline Capability. Future development may introduce an offline-capable architecture to address situations where internet connectivity is unavailable or unstable [10]. The system could temporarily store authorized transactions on the user's device or within a local application cache and automatically synchronize the stored information with the central database once connectivity is restored. This functionality would be particularly useful for attendance recording and other routine activities that need to continue during temporary network interruptions. Synchronization rules should be carefully designed to prevent duplicate records, conflicting updates, or accidental data loss. Error handling and synchronization status indicators should also be included so users can determine whether their data has been successfully uploaded to the central system.

Learning Management Extension. The TCCIS SMS may be expanded with a dedicated Learning Management System (LMS) module to support digital and blended learning activities [4]. Potential features may include lesson or learning-material distribution, assignment posting and submission, online assessment, feedback, activity tracking, and student learning progress monitoring. Integration with the existing student and teacher accounts would allow users to access these functions using their existing credentials. The extension should be designed as a separate but integrated component so that the core school management functions remain organized and easy to use. Future implementation should also consider storage requirements, content management, user accessibility, and data privacy.

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Financial Integration. Future versions may incorporate secure financial management and payment capabilities to extend the system's administrative functions [3]. A financial module could support fee records, payment monitoring, expense recording, budgeting, and financial report generation. If online payments are introduced, the system should use an appropriate secure payment gateway and follow applicable security and data-protection requirements. Payment information should not be stored unnecessarily within the school database, and access to financial information should be restricted to authorized personnel. Integration between financial records and student profiles could also reduce repetitive encoding and provide administrators with a more centralized view of school-related financial information.

Advanced Analytics. The system may be enhanced with interactive dashboards and analytical tools to provide administrators with a clearer view of important school trends [28]. Dashboards could present attendance patterns, grade distributions, inventory usage, student enrollment trends, and other relevant indicators through charts, graphs, tables, and summary cards. Instead of requiring users to examine individual reports manually, visual analytics could highlight changes and patterns over selected periods. Filters could allow administrators to view information by grade level, class, subject, school year, or date range. Such features could support data-informed planning while ensuring that interpretations remain based on accurate and appropriately protected data.

Improved Notification Features. Future development may also introduce configurable notifications for important events and system activities. Notifications could inform authorized users about prolonged student absences, low inventory levels, newly published announcements, pending reports, or other school-defined events. Notifications may be presented within the system and, where appropriate and authorized, through additional communication channels. Users should be provided with appropriate notification preferences to prevent excessive or unnecessary alerts.

Enhanced Security and Data Protection. As the system expands, additional security mechanisms should be considered to protect student, parent, teacher, and administrative information. Future improvements may include stronger password policies, multi-factor authentication for administrative accounts, automated session management, regular security assessments, database backups, and improved audit-log monitoring. Access permissions should also be reviewed periodically to ensure that users retain only the privileges required for their responsibilities. Regular backups and recovery procedures should be established to reduce the potential impact of hardware failure, accidental deletion, or other data-related incidents.

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Mobile and Accessibility Improvements. Future versions may further optimize the system for smartphones and tablets to improve accessibility for parents and teachers who primarily use mobile devices. Responsive interfaces, simplified navigation, readable layouts, and accessible form controls should be considered during development. Accessibility improvements should allow users with different levels of technical familiarity to interact with the system more easily while maintaining the security and functionality of the existing platform.

System Maintenance and Continuous Evaluation. Finally, the school should establish a regular maintenance and evaluation process after implementation. System performance, security, user feedback, database integrity, and feature usage should be monitored periodically. Software dependencies and hosting configurations should also be updated when necessary. User feedback from administrators, teachers, and parents can be collected regularly to identify new requirements and areas for improvement. Future system versions can then be developed through incremental updates, ensuring that TCCIS SMS continues to adapt to changes in school operations and technological requirements.

These recommendations provide possible directions for extending the capabilities of TCCIS SMS beyond its current implementation. Prioritization of future enhancements should consider the school's available budget, technical resources, user requirements, privacy obligations, infrastructure, and operational priorities. Through gradual and properly evaluated improvements, the system can continue to develop while maintaining its core purpose of supporting efficient, organized, secure, and accessible school management.

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